System: **系统：**	**ABS 9.x** **ESP® 9.x** **ESP®plus 9.x**	**Equipment 设备: Evacuation and filling machine 抽真空加注机** **Diagnostic connection: K-Line (opt.); CAN; FlexRay; etc. (customer specific)**
Customer: **客户：**	**General 通用**	**Diagnosis protocol 诊断协议: Customer specific 客户规范**
Distribution 发送: Customer 客户		

History list 历史列表:

No.	Date	Pages	Description	Signature
00	17.09.2008	All	New document	Sgd. A. Horvath
01	06.04.2009	See page 2	See page 2	Sgd. A. Horvath
02	01.12.2009	All	Translate into Chinese	Sgd. Li.Hunter Lin. Xiang

	Name	Signature:	Date:
Checked by:	Jiang Zhigang		
Approved by:	Przybyla Benjamin		

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Modifications修改

Version:	Modifications:
00	New document
01	Pages 14 to 16: Chapter „ Process step 2: Compressed air leakage test (optional) “ and “Evacuation cycle 1 inclusive coarse leakage test “ updated. Note to the valve activation time t_{Act} updated ($t_{Act} = t_{Actmin}$). Pages 40 to 43: Chapter „Diagnostic step 3: Start of the routine evacuation and filling to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit“ updated. Note and information to the minimum activation time t_{Actmin} and minimum cooling time $t_{Coolmin}$ added. Pages 48: Chapter „Process data (absolute pressure and time values)“ updated. The minimum activation time t_{Actmin} and minimum cooling time $t_{Coolmin}$ deleted. Page 51: Chapter „Diagnostic data (Times)“ updated. The minimum activation time t_{Actmin} and minimum cooling time $t_{Coolmin}$ added. Page 52: Chapter „Hydraulic data“ updated. Missing data added. Pages 57: Chapter „Note: Installation of a termination resistance within the CAN Bus“ added.

对 本 文 件 有 疑 问 请 以 英 文 为 准 !

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Table of contents目录:

1 GENERAL 概要 5

1.1 ACRONYMS 术语 5

2 INTRODUCTION 导言 6

3 PROCESS DESCRIPTION 过程描述 7

3.1 SCOPE 范畴 7

3.2 PROCESS SEQUENCE 过程序列 9

4 DESCRIPTION OF EACH SINGLE PROCESS STEP 每一步骤的描述 12

4.1 PROCESS STEP 1 步骤 1: HANDLING 1 操作 1 12

4.2 PROCESS STEP 2 步骤 2: COMPRESSED AIR LEAKAGE TEST (OPTIONAL) 压缩空气泄漏测试（可选） 14

4.3 PROCESS STEP 3 步骤 3: EVACUATION CYCLE 1 INCLUSIVE COARSE LEAKAGE TEST 包含泄漏测试的抽真空 17

4.4 PROCESS STEP 4 步骤 4: LEAKAGE TEST 泄漏测试 21

4.5 PROCESS STEP 5 步骤 5: EVACUATION CYCLE 2(OPTIONAL) 抽真空循环 2（可选） 23

4.6 PROCESS STEP 6 步骤 6: FILL CYCLE 加注循环 25

4.7 PROCESS STEP 7 步骤 7: PRESSURE TEST 压力测试 28

4.8 PROCESS STEP 8 步骤 8: LEVELLING OF THE BRAKE FLUID LEVEL IN THE BRAKE FLUID RESERVOIR 31

4.9 PROCESS STEP 9 步骤 9: END OF PROCESS/HANDLING 2 结束过程/操作 2 33

5 DESCRIPTION OF EACH SINGLE DIAGNOSTIC STEP 每个单独诊断步骤的描述 35

5.1 DIAGNOSTIC STEP 1 诊断步骤 1: START OF COMMUNICATION 开始通讯 38

5.2 DIAGNOSTIC STEP 2 诊断步骤 2: ECU IDENTIFICATION(OPTIONAL) 电子控制单元识别（可选） 39

5.3 DIAGNOSTIC STEP 3 诊断步骤 3: START OF THE ROUTINE „EVACUATION AND FILLING“ TO EVACUATE AND FILL THE SECONDARY CIRCUITS OF THE ABS/ESP HYDRAULIC UNIT 开始 ABS/ESP 第二回路抽真空加注 40

5.4 DIAGNOSTIC STEP 4 诊断步骤 4: EVALUATION OF THE STATUS OF THE ROUTINE „EVACUATION AND FILLING“ 评估抽真空加注 44

5.5 DIAGNOSTIC STEP 5 诊断步骤 5: WRITE PROCESS BYTES (OPTIONAL)写入过程字节（可选） 46

6 DATA 数据 48

6.1 PROCESS DATA (ABSOLUTE PRESSURE AND TIME VALUES) 过程数据（绝对压力和时间值） 48

6.2 DIAGNOSTIC DATA (TIMES) 诊断数据（时间） 51

6.3 HYDRAULIC DATA 液压数据 52

6.4 ELECTRICAL DATA 电气数据 53

6.4.1 Information to the calculated current consumption of an ABS/ESP system 计算系统电流消耗 53

6.4.2 Specification of the voltage supply 电压规格 53

6.4.3 Specification of the ECU connection 电子控制单元连接规范 55

6.4.4 Note „Installation of a termination resistance within the CAN Bus“注意“CAN 终端电阻的安装”。 57

7 VALVE PROTECTION MEASURES AT GENERATION 9 9 代系统的阀保护措施 60

8 FAULT HANDLING 错误处理 62

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

9 ELECTRICAL ADAPTATIONS 电气连接 63

9.1 CONNECTION AT THE DIAGNOSTIC CONNECTOR (OBD2) WITHIN THE VEHICLE 连接车内 OBD2 接口 63

9.2 CONNECTION AT THE ECU PLUG 连接 ECU 插头 65

10 EQUIPMENT INFORMATION 设备信息 67

11 ATTACHMENTS 附件 68

11.1 HYDRAULIC CIRCUIT DIAGRAMS 液压回路图 68

11.1.1 ABS 9 - independent of the brake split configuration ABS 9-适用于所有回路布置方案 68

11.1.2 ESP 9 - independent of the brake split configuration ESP 9-适用于所有回路布置方案 68

11.2 ELECTRICAL WIRING DIAGRAMS 电路图 69

11.2.1 Electrical wiring diagram ESPi 9 ESPi9 的电路图 69

11.2.2 38-pin connector system 9 9 系统的 38 针连接器 70

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

1 General

1.1 Acronyms

ABS	antilock braking system 防抱死系统	MCC	master cylinder circuit 主缸回路
ABS-SILA	ABS warning light ABS 报警灯	PATA	passive taster (TCS/DSC passive taster)ESP 功能关闭开关
AC	accumulator chamber 蓄能器	PB	process byte 过程字节
AV	outlet valve 出液阀	PS (DG, DS)	pressure sensor 压力传感器
ALS	longitudinal acceleration sensor 纵向加速度传感器	RA	rear axle 后轴
AYS	lateral acceleration sensor 横向加速度传感器	RFP	recirculation pump 回流泵
BLS	brake light switch 制动灯开关	RL	rear left 左后
BS	brake switch 制动开关	RR	rear right 右后
DTC	diagnostic trouble code 诊断故障码	SAS (LWS)	steering angle sensor 方向盘转角传感器
EBD	electronic brake distribution 电子制动力分配	SV	solenoid valve 电磁阀
ECU	electronic control unit 电子控制单元	UMR	pump motor feedback 泵马达反馈
ESP	electronic stability program 电子稳定程序	USV	pilot valve 回路控制阀
ESP-MFL	ESP multifunction (warning) light ESP 故障报警灯	UVR	valve relay feedback 阀继电器反馈
EV	inlet valve 进液阀	WSS	wheel speed sensor 轮速传感器
FA	front axle 前轴	YRS	yaw rate sensor 横摆角速度传感器
FCM	fault code memory 故障码存储器		
FL	front left 左前		
FR	front right 右前		
HAS	handbrake switch (park brake switch)手刹开关		
HSV	high pressure switching valve (priming valve)高压开关阀（吸入阀）		

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

2 Introduction 导言

This document contains the procedure that has to be carried out to evacuate and fill dry ABS and ESP systems on line with existing equipment's and additional actuation units.

本文件描述了在抽真空加注设备上对干式 ABS 和 ESP 加注的过程.

The procedure is based on the customer specific diagnosis and the corresponding features.

本程序是建立在客户诊断规范和通讯协议的基础上.

Requirements for the test facilities provided by the customer 对客户抽真空加注设备的要求:

- Communication with the Bosch ECU 和博世的电子控制单元通讯.
- Control of the process sequence 控制过程序列
- Control of the evacuation and filling equipment 控制抽真空加注设备
- Evaluation of the data 评估数据
- Issue process protocol 发布过程协议

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

3 Process description 过程描述

3.1 Scope 范畴

All the Bosch ABS and ESP components, such as hydraulic modulator respectively the pump motor are delivered to the customer assembly plants in a dry state (free of brake fluid).
Both circuits (primary and secondary) of the hydraulic modulator are dry.
In order to evacuate and fill the entire modulator, valves must be energized. A pump motor activation is only required at the end of the filling process to discharge the accumulator chambers.
The valves and the pump motor are activated by using of an electrical adaptation and diagnosis tool. In addition an external power supply is required to supply the ABS respectively ESP ECU and the pump motor (depending on the electrical connection).
所有的博世 ABS/ESP 部件，例如液压单元和泵马达都是以干燥的状态（无制动液）提交到客户装配工厂的，其中液压单元第一回路和第二回路都是干燥的。为了对整个液压单元抽真空/加注，利用诊断设备和电气接头对阀和泵马达通电，其中泵马达通电仅在加注过程结束后进行，以排空蓄能器。此外还需要一个外接电源来为 ABS/ESP ECU 和泵马达供电（取决于电气连接）。

The procedure will then be executed in a same way than a standard foundation brake system with the exception of the evacuation and fill times to be adapted onto the design of the ABS or ESP modulator. Caused by these small orifice sizes the evacuation and fill times are extended by physical limitations (reduced volume flow) in comparison to a vehicle without ABS/ESP unit.
The electrical adaptation can either be done by connecting onto the diagnostic link of the vehicle (requires a complete installation) or onto the modulator connector (requires an external power supply).
ABS/ESP 的抽真空加注流程执行方式与标准的基础制动系统相同，唯一的差别就在在抽真空加注时间上，这取决于 ABS/ESP 液压单元的设计。由于液压单元自身截流口孔径大小限制了液体流量，这将延长抽真空和加注时间。电气接头可以连接在车身的诊断接口上（ABS/ESP 需完整安装到车上），也可以连接到 ABS/ESP 液压单元的接头上（需要外接电源供电以作为缓冲器-博世推荐）。

Technical data for external power supply are also specified in this document.
All data for an evacuation and filling process (absolute pressure (vacuum) values, minimum fill pressure, timings, etc.) which are required to evacuate and fill a foundation brake system inclusive a ABS/ESP HU are specified in this document.

The evac/fill equipment must be capable to meet the requirements as specified in this document.
本文档还规定了外接电源的技术数据，抽真空/加注设备必须能满足本文档规定的要求。
液压单元的真空度、最小加注压力数据和时间安排等在液压单元的客户技术文件中和本文档中有详细规定。

In order to achieve a sufficient vacuum level, all components, such as master cylinder, wheel callipers and pressure relieve valves must meet the minimum requirements as specified in this document.

The final timing of the production cycle can only be determined by trials with production parts (or parts close to production level) and on line equipment.
On line equipment may need to be modified onto the requirements of the hydraulic configuration of the vehicle.
为了达到一个充分的真空度，所有的元件如主缸、制动轮缸和泄压阀必须满足本文档规定的最低要求。
生产线上的最终时间安排只能由生产件（或接近生产件水平的部件）和在线设备通过试验来确定，在线设备可能根据车辆液压配置的要求进行修改。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

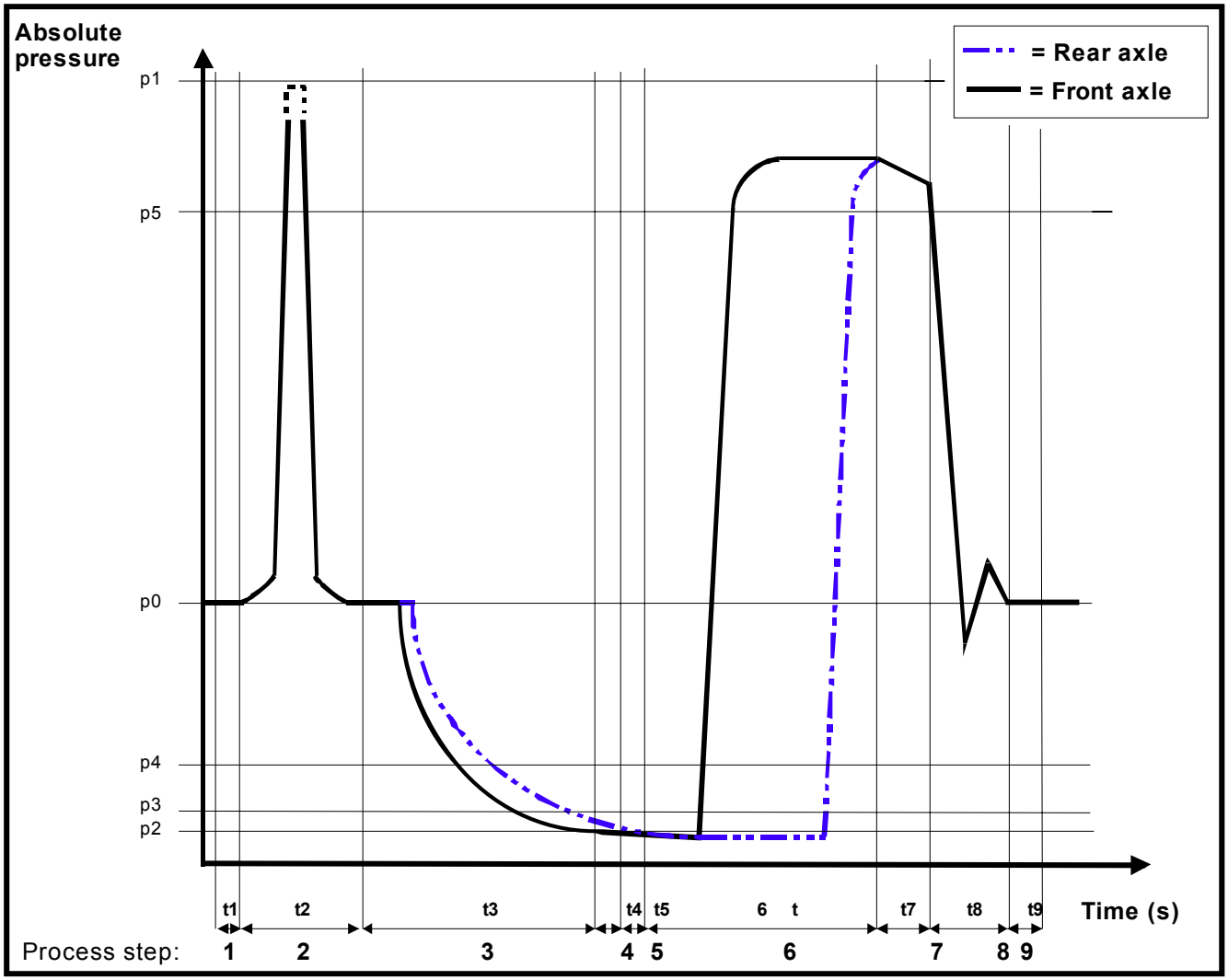
In case of any failures of the evac/fill cycle or a replacement of the hydraulic modulator, a repair bleed procedure must be performed according to a specific sequence. This special sequence is described in a special document.
假如车辆抽真空/加注失败或需要更换液压单元，必须按照指定程序进行人工排气。该指定程序请参考相关文件。

0 265 EA6 346

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

3.2 Process sequence 过程序列

Diagram of the process sequence:过程序列示意图：



Details to the diagram of the process sequence:示意图解析：

Process step:	Time:	Description of the process:
1	t1	At the beginning of the process the fill head and the electric connections are adapted. The required **time t1** for the process step **handling 1** depends on the operator.抽真空加注前加注头和电气连接应当安装好，操作 1 所需时间 t1 由操作员决定。
2	t2	The optional compressed air leakage test is only allowed before the evacuation cycle. Appropriate process parameters must be kept. **The test time t2** depends on the test parameters and the whole brake system. 压缩空气泄漏测试（可选）仅在抽真空循环前进行。适当的过程参数应予以保留。时间 t2 由测试参数和整个制动系统决定。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

3	**t3**	The **evacuation cycle 1** is required to evacuate the brake system. To detect a coarse leakage, a coarse leakage test is executed after a defined waiting period. It is evacuated as long as, since a defined minimal pressure value is reached in the wheel callipers. For evacuating the rear wheel callipers a longer time is required compared to the front ones. Caused by the used orifices (small diameters) in the ABS or ESP hydraulic unit the **evacuation time t3** is longer than at vehicles with conventional brake system. Therefore the whole time of the evacuation and fill process is affected by the required diagnosis functions only insignificantly. The activation of the ABS or ESP valves is necessary to evacuate the secondary circuit of the hydraulic unit. Details to the required valve activations see in the chapter Diagnostic step 3: Start of the routine „evacuation and filling“ to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit. 抽真空循环 1 对整个制动系统抽真空，直到制动轮缸内达到规定的最小压力值，后轴抽真空所需时间比前轴的长，带 ABS 或 ESP 的车比不带的时间长。为了检测是否存在较大的泄漏，在规定的等待时间内进行大泄漏测试。 由此可见，诊断功能耗时对整个抽真空加注过程耗时影响不大。ABS 或 ESP 阀的启动对液压单元第二回路抽真空至关重要，具体请参考“诊断步骤 3”。
4	**t4**	The vacuum leakage test is required to test the low pressure tightness of all screw connections and the whole brake system. The **leakage test time t4** is a time which depends on the evac/fill machine. It is not influenced by the ABS or ESP unit. 真空泄漏测试用来确保所有排气螺钉连接和整个制动系统的低压气密性良好。时间 t4 由抽真空加注设备决定，不受 ABS 或 ESP 单元影响。
5	**t5**	The **evacuation cycle 2** (optional) is required to reach the initial pressure after the vacuum leakage test. The required **evacuation time t5** is a time which depends on the evac/fill machine. It is not influenced by the ABS or ESP unit. 在真空泄漏测试结束后，抽真空循环 2（可选）应使系统达到初始压力，时间 t5 由抽真空加注设备决定，不受 ABS 或 ESP 单元影响。
6	**t6**	The **fill cycle** is required to fill the whole brake system. To fill the rear brake callipers takes longer are than the front caused by the bigger volumes and lower orifice diameter. The **filling time t6** is the time which is demanded to reach the specified pressure value in the brake callipers. The **activation** of the corresponding **ABS / ESP valves** is necessary **to fill the secondary circuits** of the hydraulic unit. Details to the required valve activations see in the chapter Diagnostic step 3: Start of the routine „evacuation and filling“ to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit. 加注循环应对整个制动系统进行加注。由于后轴制动轮缸容积更大，截流口孔径更小，因此加注时间要比前轴长。加注时间 t6 是在制动轮缸中达到规定压力所需的时间。加注第二回路时，ABS/ESP 相应阀需要启动，具体请参考“诊断步骤 3”。
7	**t7**	The **pressure test** after the filling process is required to test the high pressure tightness of all screw connections and the whole brake system. The **pressure test time t7** is a time which depends on the evac/fill machine. It is not influenced by the ABS or ESP unit.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

		加注后的压力测试用来确保所有排气螺钉连接和整个制动系统的高压气密性完好。时间 t7 由抽真空加注设备决定，不受 ABS 或 ESP 单元影响。
8	**t8**	The **levelling of the brake fluid state in the reservoir** is required to adjust the correct brake fluid level. The **time t8** is demanded to reduce the pressure in the reservoir to atmospheric pressure and to put the brake fluid state in the reservoir on the correct value. This time is a time which depends on the evac/fill machine. It is not influenced by the ABS or ESP unit.时间 t8 用来使储液壶中的压力下降到环境大气压并使壶内制动液面达到正确位置。该时间由抽真空加注设备决定，不受 ABS 或 ESP 单元影响。
9	**t9**	At the end of the entire process the fill head and the electric connections are separated. The required **time t9** for the **handling 2** depends on the operator. 抽真空加注整个循环结束后加注头和电气连接应当拆除，操作 2 所需时间 t9 由操作员决定。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

4 Description of each single process step 每一步骤的描述

4.1 Process step 1 步骤 1: Handling 1 操作 1

Handling 1	
Operator:	**Evacuation and filling equipment/ ECU:**
☞ **Adaptation of the fill head at the brake fluid reservoir** 将加注头安装在储液壶上 ☞ **Adaptation of all electrical connections** 安装好所有电气连接 ☞ **Switching on of the ignition (required in case of connection via the OBD2 diagnostic interface)** **Switching on of the external power supply (required in case of direct connection at the ABS or ESP ECU)** 打开点火开关（适用于通过 **OBD2** 诊断接口连接） 打开外接电源（适用于通过 **ABS/ESP ECU** 直接连接） ☞ **Starting of the process** 开始过程	**Switching on of the external power supply (direct connection)** 打开外接电源（直接连接时）

Process step:	Parameter:	Physical values:
1	Handling time t_1	x s

Notes注意:

The with x marked handling time t1 depends on the installation position of the brake fluid reservoir, the hydraulic unit respectively the position of the diagnosis interface OBD2 in the vehicle interior. The handling time t1 must be determined together with the operators by trials. These trials are in responsibility of the customer and have to be executed by them self.
数值为 x 的操作时间 t1 由储液壶的位置、液压单元的位置以及 OBD2 诊断接口的位置决定，需要客户通过试验确定。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

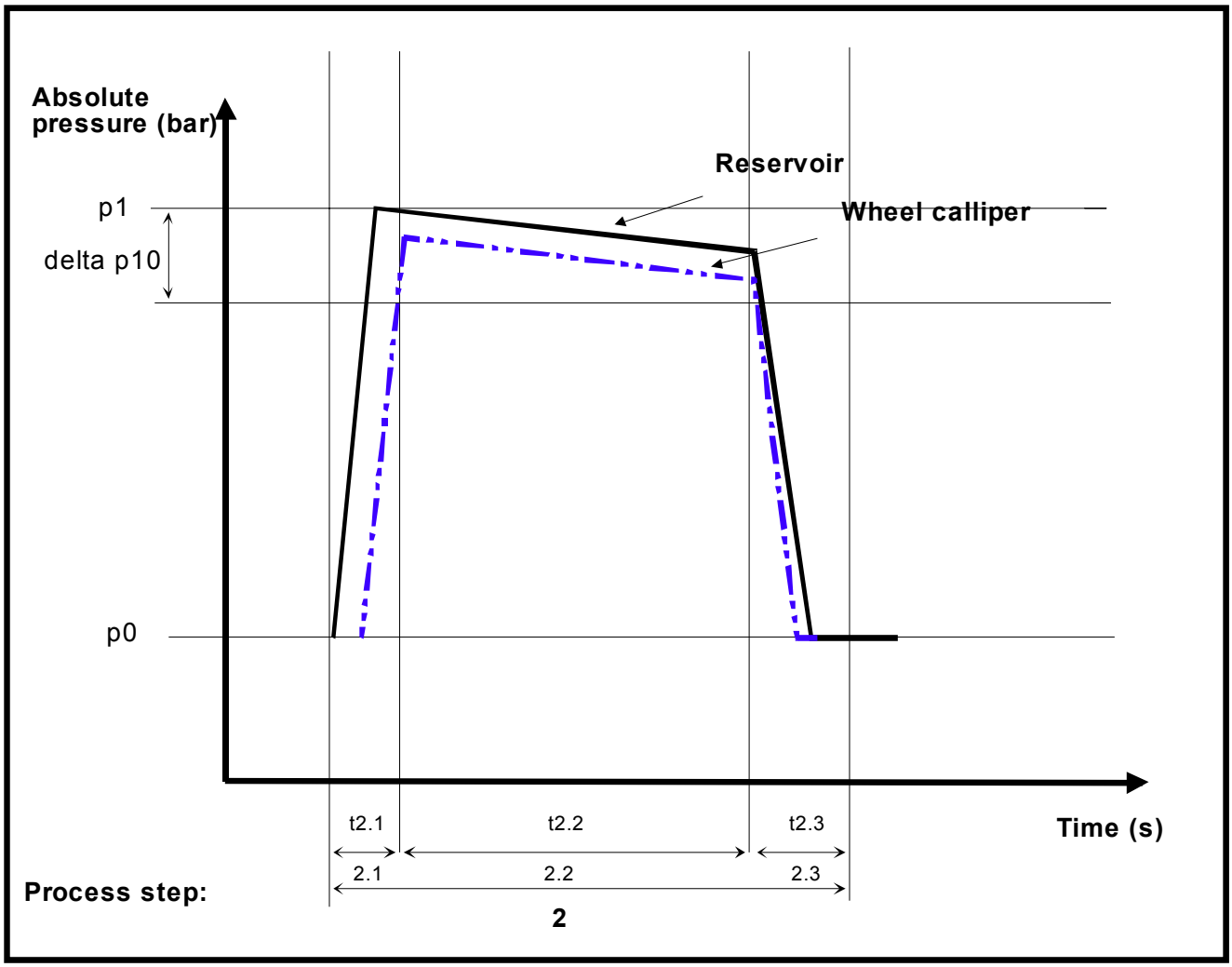
The electric adaptation can be done either by the diagnosis interface of the vehicle (OBD2) or directly via the ECU connector (according to the design of the evac/fill process, the accessibility to the ABS or ESP unit and the installation conditions of the wiring harness in the vehicle. Details to the possible electrical adaptations can be inferred from the chapter Electrical adaptations.
电气接头可以连接到 OBD2 或电子控制单元的连接器上（根据抽真空加注流程设计、ABS/ESP 是否容易接触以及车辆线束安装条件来决定，具体请参考“电气接头”相关章节）。

According to the electrical adaptation either the operator must turn on the ignition or the evac/fill machine must turn on the external power supply.
根据电气接头连接方式的不同，操作员必须打开点火开关或打开外接电源。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

4.2 Process step 2 步骤 2: Compressed air leakage test (optional) 压缩空气泄漏测试（可选）

Diagram of the „compressed air leakage test“压缩空气泄漏测试示意图:



Description to the „compressed air leakage test“泄漏测试描述:

Process step:	Time:	Process description:
2.1	$t_{2.1}$	**Build up und stabilisation of the test pressure p_1** within the brake system.测试压力p_1上升及稳定
2.2	$t_{2.2}$	Measurement of the initial and final pressure during the **test time $t_{2.2}$** and evaluation of the occurred **pressure decay Δp_{10}.** 测量$t_{2.2}$前后压力，评估压力减小量Δp_{10}
2.3	$t_{2.3}$	**Pressure release** to **ambient-air pressure p_0**.压力下降至环境大气压p_0

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Process parameter过程参数:

Process step:	Test parameter:	Physical values:	Test medium:
2.1	**Pressure build up time $t_{2.1}$:** 压力上升时间**$t_{2.1}$**	**x s**	**Air (dry and free of oil); nitrogen** 干燥无油的空气；氮气
	Maximal absolute **pressure value p_1:** 最大绝对压力值**p_1**	**y bar**	
2.2	Maximal allowed **pressure decay Δp_{10}:**最大允许压力减小值**Δp_{10}**	**x mbar**	
	which is allowed at the brake fluid reservoir for the **test time $t_{2.2}$:** 测试时间**$t_{2.2}$**	**x s**	
2.3	**Pressure release time $t_{2.3}$:** 压力下降时间**$t_{2.3}$**	**x s**	
	Atmospheric **ambient-air pressure p_0:**环境大气压**p_0**	**~ y bar**	

Notes注意:

All with x marked time and pressure values depend on the used brake components (e.g. a maximum test pressure depends on the brake fluid reservoir, etc), evac/fill equipment and the maximal available cycle time. Therefore the **entire time t2 (consisting of $t_{2.1}$, $t_{2.2}$ and $t_{2.3}$)** must be defined by the customer.
数值为 x 的所有时间和压力值取决于所采用的制动元件（如最大测试压力取决于储液壶）、抽真空加注设备和最大可用循环时间。因此整体时间 t2 应由客户定义。
All with y marked time or pressure values are to be inferred from the chapter Process data (absolute pressure and time values).
数值为 y 的所有时间或压力值可由“过程数据（绝对压力和时间值）”相关章节中得出。
The optional compressed air test was designed to check the tightness of the whole brake system before the evacuation process. This test affects not the general evacuation and filling process, however it is an additional test step.
压缩空气泄漏测试（可选）用来在抽真空之前测试整个制动系统的气密性。作为一个额外的测试步骤，该测试不影响整个抽真空加注循环。
The evacuation and filling equipment switches off the pressure source after reaching of the initial **pressure value p_1** (the **maximum allowed pressure value p_1 depends on the design of the brake fluid reservoir**) and a defined stabilization time (contained in the pressure time $t_{2.1}$), then it measures after the **test time $t_{2.2}$** the final pressure $p_{1.1}$. Afterwards the evacuation and filling equipment compares the measured values and compare them with the given tolerances (maximally allowed **pressure decay Δp_{10}**). After finishing of this test the pressure is released to **ambient-air pressure p_0**.
在达到初始压力值**p_1**（**p_1**最大允许值取决于储液壶的设计）且经过一个规定的稳定时间（包含在$t_{2.1}$中）之后，抽真空加注设备切断压力源，并在**$t_{2.2}$**后测量最终压力$p_{1.1}$。随后抽真空和加注设备对比前后压力值，并判断其差值是否处于允许范围内（最大不超过**Δp_{10}**）。在该测试结束后，减小压力至环境大气压**p_0**。
As test medium air (dried and free of oil) or nitrogen can be used.

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

干燥无油的空气或氮气可用做测试介质。

The maximally permissible pressure value p_1 must be asked at the manufacturer of the brake fluid reservoir.

p_1最大允许值应向储液壶制造商确认。

The ABS / ESP valves and the pumping engine may not be activated during the entire test!

ABS/ESP 的阀组和泵马达在整个测试期间不能启动。

At the following evacuation process must be guaranteed that the ABS / ESP valves are activated for a defined valve activation time t_{Actmin}. This is necessary, because caused by the defined and allowed leakage value of the AV's (specified in the HU TCD), air can enter into the accumulator chamber or pumping elements. To evacuate this compressed air certainly from the brake system, the activation of the valves for this minimum valve activation time t_{Actmin} is absolute required.

Details to the required valve activation see in the chapter Diagnostic step 3: Start of the routine „evacuation and filling“ to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit.

在下面的抽真空过程中，必须保证ABS/ESP阀组在规定时间内**t_{Actmin}**内启动。这是因为AV阀（详见液压单元客户技术文件）有一定的允许泄漏值，空气可以由此进入蓄能器或泵单元。为了将压缩空气从制动系统中排空，阀最短启动时间t_{Actmin}是必不可少的。

阀启动具体要求请参考“诊断步骤 3”。

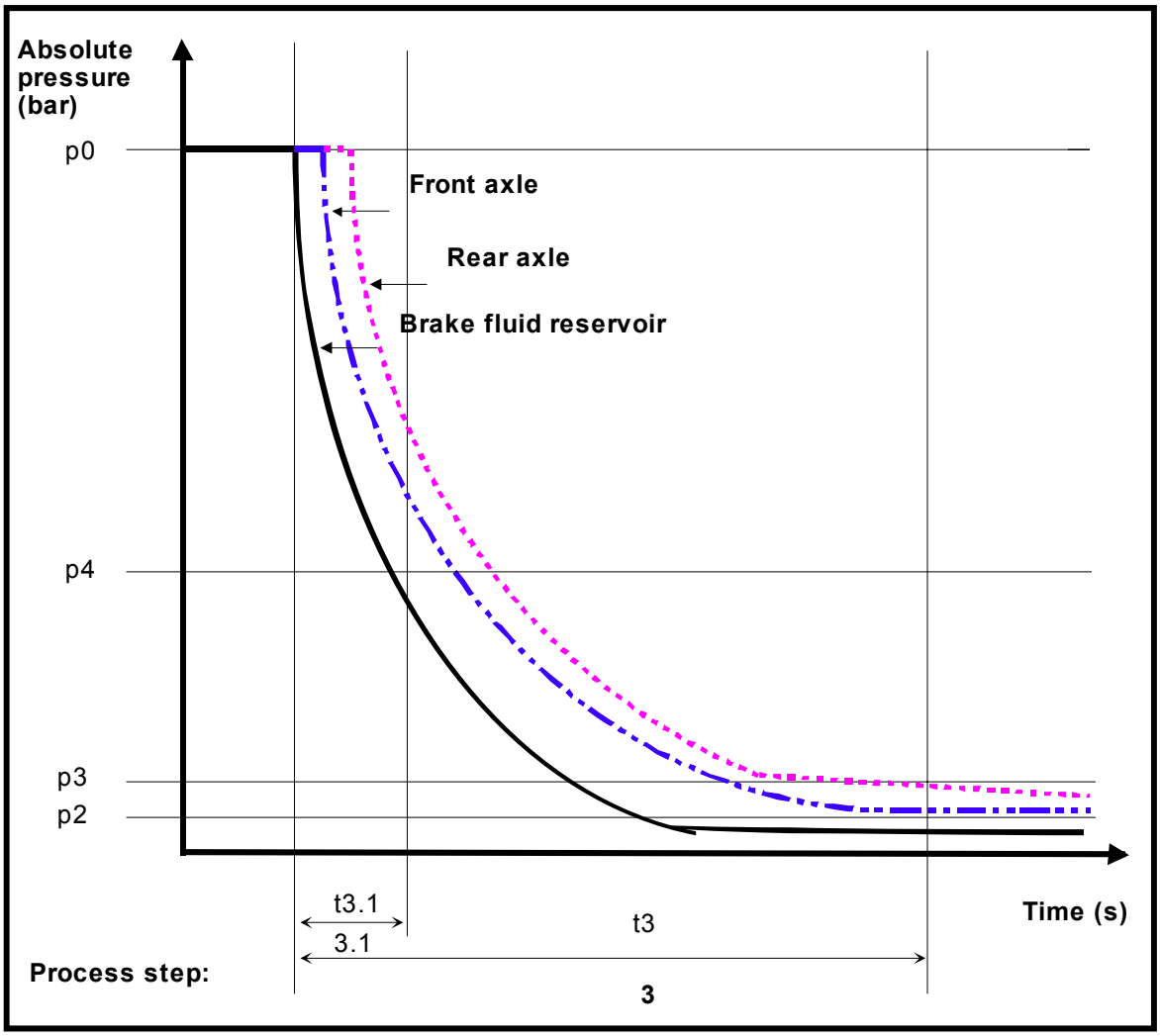
Compressed air leakage test ok 压缩空气泄漏测试成功 ⇒ **Continue** 继续

Compressed air leakage test nok 压缩空气泄漏测试不成功 ⇒ **Abort, print out fault message** 退出，打印错误信息

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

4.3 Process step 3: Evacuation cycle 1 inclusive coarse leakage test
步骤 3：包含大泄漏测试的抽真空循环 1。

Diagram „ Evacuation cycle 1 inclusive coarse leakage test“:



Details to the diagram „ Evacuation cycle 1 inclusive coarse leakage test“:
示意图解析：包含大泄漏测试的抽真空循环 1

Process step:	Time:	Process description:
3	t_3	During the **evacuation cycle 1** the whole brake system is evacuated including the secondary circuits of the ABS / ESP hydraulic unit (activation of the corresponding ABS / ESP valves necessary) to the given **pressure values p_2 and p_3**. The whole **evacuation time t_3** is to be investigated by the customer.在抽真空循环 1 内整个制动系统包括ABS/ESP液压单元第二回路在内（ABS/ESP相应阀组需要启动）都被抽真空至规定压力值**p_2**与**p_3**。整个抽真空时间**t_3**由客户进行确认。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

3.1	$t_{3.1}$	The coarse leakage test detects coarse leakages within the screw connections and components of the whole brake system. The minimum expected **pressure value p_4** and the **delay time $t_{3.1}$** are to be investigated by the customer.大泄漏测试用来检测排气螺钉连接和整个制动系统部件中是否存在较大泄漏。预计的最小压力值**p_4**和延迟时间**$t_{3.1}$**由客户进行确认。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Process parameter过程参数:

Process step:	Test parameter:	Physical values:
3	Minimum expected absolute **pressure value p_2** at the **brake fluid reservoir**:储液壶中的预计最小绝对压力值**p_2**	**≤ y mbar**
	Minimum expected absolute **pressure value p_3** at the **wheel callipers**:制动轮缸中的预计最小绝对压力值**p_3**	**≤ y mbar**
	Evacuation time t_3抽真空时间**t_3**	**x s**
3.1	Minimum expected absolute **pressure value p_4** at the **brake fluid reservoir**: 储液壶中的预计最小绝对压力值**p_4**	**≤ y mbar**
	after a **delay time $t_{3.1}$** and start of the evacuation cycle 1: 延迟时间**$t_{3.1}$**	**x s**

Notes注意:

The with x marked **evacuation time t_3** and the **delay time $t_{3.1}$** must be determined by trials with components close to series and the production equipment. These trials have to be executed by the customer.
数值为 x 的抽真空时间和延迟时间必须由接近生产件水平的部件和生产设备通过试验来确定。这些试验必须由客户来进行。
All with y marked time and pressure values are to be inferred from the chapter Process data (absolute pressure and time values).
数值为 y 的时间和压力值可由“过程数据（绝对压力和时间值）”相关章节中得出。
The vacuum filling equipment must be able to reach a sufficient vacuum value in the brake fluid reservoir respectively whole brake system.
The time to reach the defined **pressure value p_2** in the brake fluid reservoir respectively **the pressure value p_3** in the wheel brake callipers is a function of the volumes of the whole brake system, the wheel brake callipers, the pipes or hoses, the size of the orifices of the ABS / ESP hydraulic unit and the suction power of the vacuum pump. During the evacuation cycle 1 the front axle will reach the **pressure value p_3** before the rear axle, because of the bigger orifice diameters in the ABS / ESP hydraulic unit.
The time to reach the pressure value p_3 at the rear axle defines the **evacuation time t_3**.
The required evacuation time t_3 for the production cycle can only be determined by trials with production parts or parts which are close to the series. These trials must be done with the series equipment.
抽真空加注设备应能在储液壶中以及整个制动系统中实现充分的真空度。储液壶中达到规定压力值**p_2**和制动轮缸中达到压力值**p_3**所需的时间与整个制动系统容积、制动轮缸、制动管路或软管、液压单元截流口孔径以及抽真空泵功率都有关系。在抽真空循环 1 中，前轴将先于后轴达到压力值**p_3**，这是因为前轴对应的液压单元孔径更大。
后轴制动轮缸达到压力值**p_3**所需的时间决定了抽真空时间**t_3**。
生产线上的抽真空时间**t_3**只能由生产件或接近生产件水平的部件通过试验来确定。这些试验必须通过生产设备来进行。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

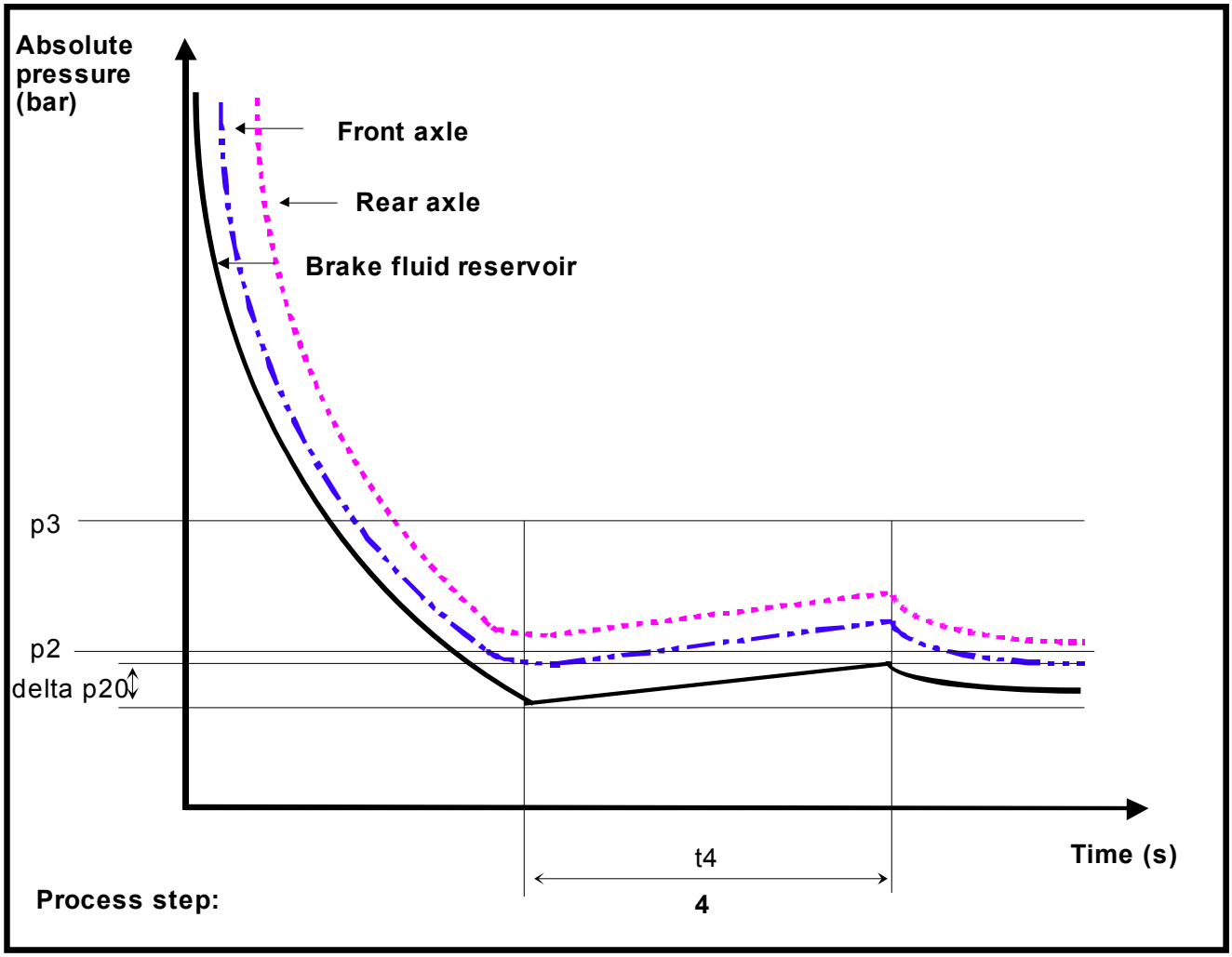
The **coarse leakage test** detects coarse leakages within the screw connections and components of the whole brake system. After starting of the evacuation cycle1 the pressure value p_4 must be reached after a defined **delay time $t_{3.1}$**. This pressure value depends on several factors and must be determined by trials.
In case of a fault the entire process must be interrupted and a fault message must be indicated.
大泄漏测试用来检测排气螺钉连接和整个制动系统部件中是否存在较大泄漏。在抽真空循环 1 开始后，在规定的延迟时间**$t_{3.1}$**内必须达到压力值**p_4**。该压力值取决于多种因素，必须由试验确定。
如果测试失败，必须中断整个过程并显示错误信息。
An **activation** of the **ABS / ESP valves** is necessary **to evacuate the secondary circuits of the hydraulic unit**. Details to the required valve activation see in the chapter Diagnostic step 3: Start of the routine „evacuation and filling“ to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit.
为了排空液压单元第二回路，必须启动 ABS/ESP 相应阀组。具体请参考“诊断步骤 3”相关章节。
All necessary diagnosis steps (details see in the associated chapter Description to the separate diagnosis steps) can be integrated into the first seconds of the evacuation. Therefore only a minimal process time continuation arises.
所有诊断步骤（具体请参考相关诊断步骤章节）都可以集成在抽真空过程开始的一小段时间内进行，从而实现整个过程时间的最小化。
After an executed compressed air leakage test must be guaranteed that the ABS / ESP valves are activated for a defined valve activation time t_{Actmin}. This is necessary, because caused by the defined and allowed leakage value of the AV's (specified in the HU TCD), air can enter into the accumulator chamber or pumping elements. To evacuate this compressed air certainly from the brake system, the activation of the valves for this minimum valve activation time t_{Actmin} is absolute required.
在压缩空气泄漏测试结束后，必须保证ABS/ESP阀组在规定时间内**t_{Actmin}**内启动。这是因为AV阀（详见液压单元客户技术文件）有一定的允许泄漏值，空气可以由此进入蓄能器或泵单元。为了将压缩空气从制动系统中排空，阀最短启动时间t_{Actmin}是必不可少的。

Evacuation cycle 1 and coarse leakage test ok **抽真空循环 1 和大泄漏测试合格**	⇒ **Continue** ⇒继续
Evacuation cycle 1 and coarse leakage test nok **抽真空循环 1 和大泄漏测试不合格**	⇒ **Abort, print out fault message** ⇒退出，打印错误信息

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

4.4 Process step 4 步骤 4: Leakage test 泄漏测试

Diagram „Leakage test“:



Details to the diagram „Leakage test“: “泄漏测试”示意图解析

Process step:	Time:	Process description:
4	t_4	The **leakage test** detects vacuum leakage within the screw connections and components of the whole brake system. Measurement of the initial and final pressure during the **test time t_4** and evaluation of the occurred **pressure increase Δp_{20}.** 泄漏测试检测排气螺钉连接和整个制动系统部件的真空泄漏。测量**t_4**前后压力，评估压力增加量**Δp_{20}**。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Process parameter:过程参数

Process step:	Test parameter:	Physical values:
4	Maximal allowed **pressure increase Δp_{20}:** 最大允许压力增加值**Δp_{20}**	**x mbar**
	which is allowed at the brake fluid reservoir for the **test time t_4:**测试时间**t_4**	**x s**

Notes 注意:

All with x marked pressure and time values depend on the used brake components, evac/fill equipment and the maximal available cycle time. Therefore the **test time t_4** and the permissible **pressure increase Δp_{20} must** be defined by the customer.
数值为x的压力和时间值取决于所用制动元件、抽真空加注设备和最大可用循环时间。因此，测试时间**t_4**和压力增加允许值**Δp_{20}**必须由客户定义。
The leakage test should detect small leakages within the whole brake system.
泄漏测试应能检测到整个制动系统中的微小泄漏。

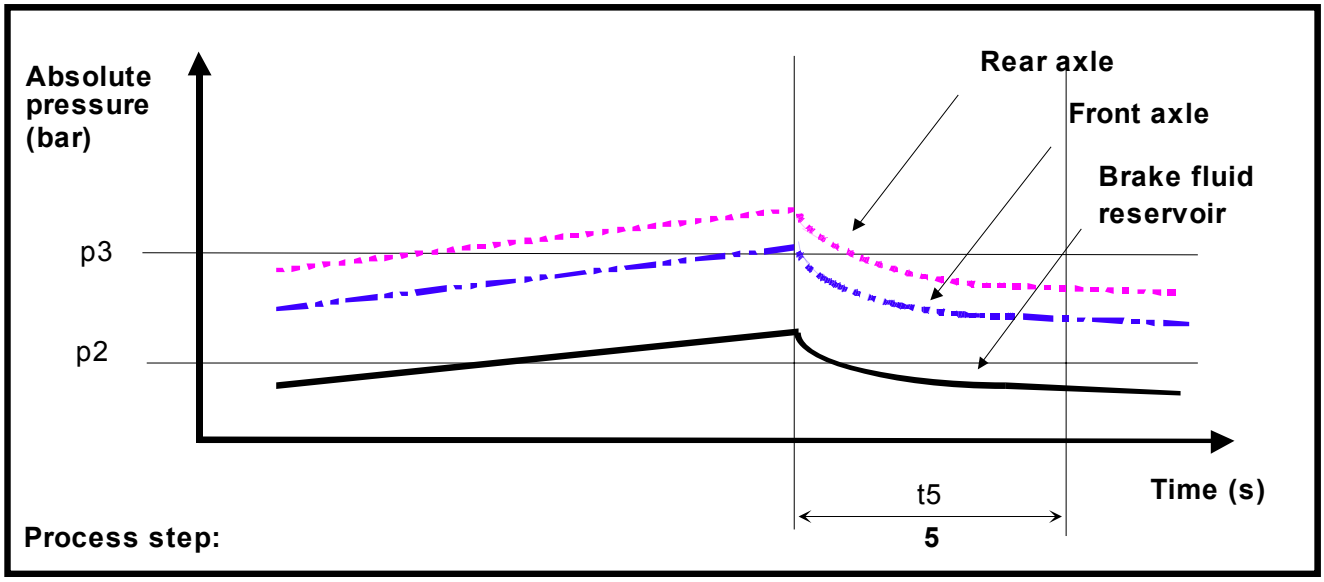
The evacuation and filling equipment isolates the evacuation pump, measures the **pressure increase Δp_{20}** within the defined **test time t_4** and compares this value with the given limits.
抽真空加注设备断开抽真空泵，测量规定时间**t_4**内的压力增加值**Δp_{20}**，并判断是否在允许范围内。

Leakage test ok 泄漏测试合格 ⇒ **Continue** 继续
Leakage test nok 泄漏测试不合格 ⇒ **Abort, print out fault message**
退出，打印错误信息

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

4.5 Process step 5 步骤 5: Evacuation cycle 2(optional) 抽真空循环 2（可选）

Diagram „Evacuation cycle 2“:



Details to the diagram „Evacuation cycle 2“:示意图解析“抽真空循环 2”

Process step:	Time:	Process description:
5	**t_5**	During the **evacuation cycle 2** the initial pressure value (≤ **pressure value p_2** in the brake fluid reservoir or ≤ **pressure value p_3** in the wheel brake callipers) is re-established. The whole **evacuation time t_5** is to be investigated by the customer.在抽真空循环 2 中，系统重新恢复到初始压力值（储液壶中压力≤ **p_2**或制动轮缸中压力≤ **p_3**）。整个抽真空时间**t_5**由客户进行确认。

Process parameter 过程参数:

Process step:	Test parameter:	Physical values:
5	To re-establish absolute **pressure value p_2** at the brake fluid reservoir:恢复储液壶中绝对压力值至**p_2**	**≤ y mbar**
	To re-establish absolute **pressure value p_3** at the wheel callipers: 恢复制动轮缸中绝对压力值至**p_3**	**≤ y mbar**
	Evacuation time t_5抽真空时间**t_5**	**x s**

Notes 注意:

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

The with x marked time value depends on the evac/fill equipment and the maximal available cycle time. Therefore the **evacuation time t5** must be determined by the customer.
数值为x的时间值取决于抽真空加注设备和最大可用循环时间。因此抽真空时间t_5必须由客户确认。
All with y marked time and pressure values are to be inferred from the chapter Process data (absolute pressure and time values).
数值为 y 的所有时间和压力值可由“过程数据（绝对压力和时间值）”相关章节中得出。
The needed time for this step has to be added to the general evacuation requirements of the whole brake system.
该步骤所需时间应添加到整个制动系统的总体抽真空时间之中。

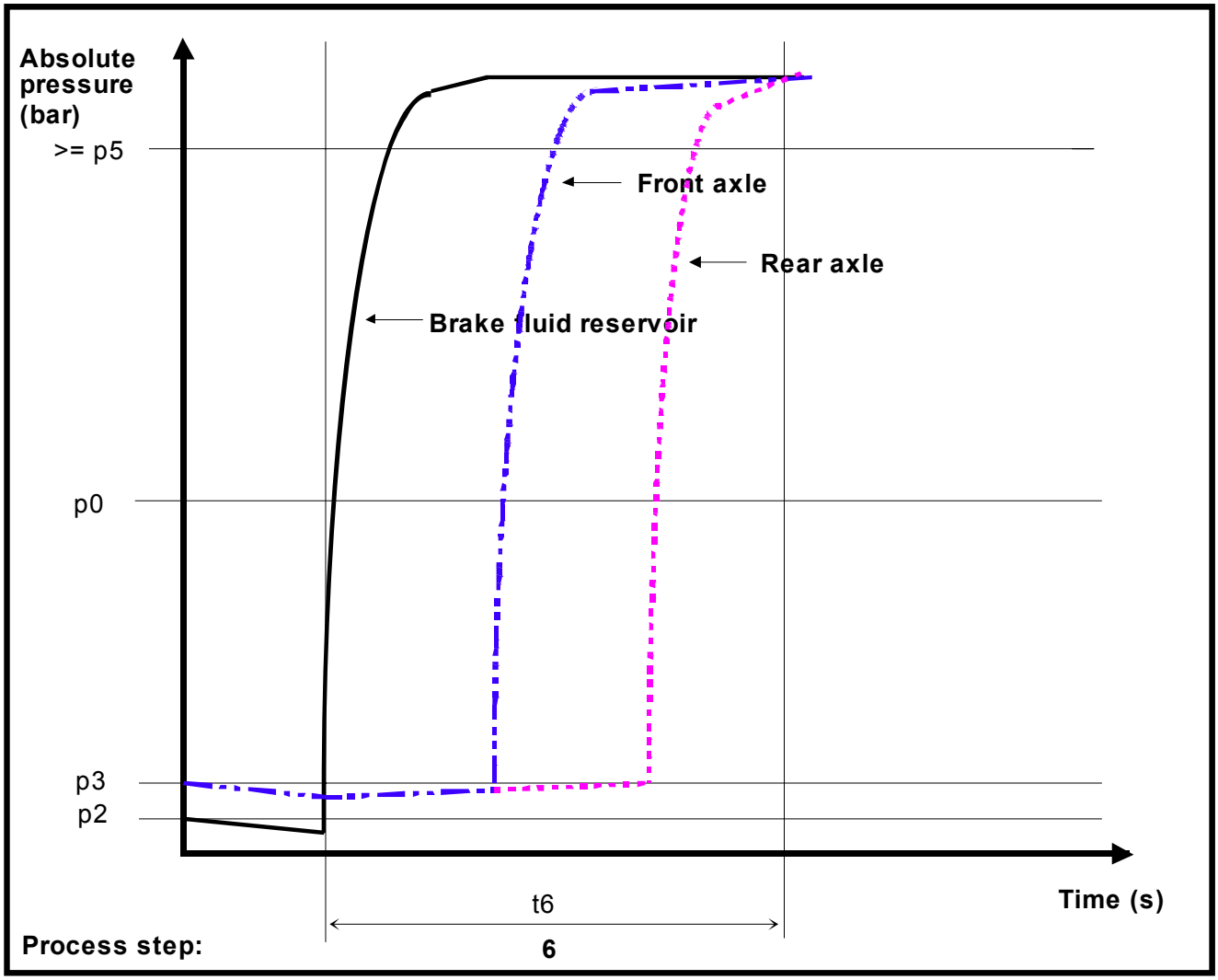
The **evacuation cycle 2** is executed for the evacuation time t5 after the leakage test to reach the initial pressure value ≤ **pressure value p_2** in the brake fluid reservoir respectively ≤ **pressure value p_3** in the wheel brake callipers.
在泄漏测试后进行的耗时t_5的抽真空循环 2 中，系统重新恢复到初始压力值（储液壶中压力≤ p_2且制动轮缸中压力≤ p_3）。

Evacuation cycle 2 ok	抽真空循环 2 合格	⇒	**Continue 继续**
Evacuation cycle 2 nok	抽真空循环 2 不合格	⇒	**Abort, print out fault message** 退出，打印错误信息

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

4.6 Process step 6 步骤 6: Fill cycle 加注循环

Diagram „Fill cycle“:



Details to the diagram „Fill cycle“:示意图解析“加注循环”

Process step:	Time	Process description
6	t_6	During the **fill cycle** the whole brake system including the secondary circuits in the ABS / ESP hydraulic unit (activation of the corresponding ABS / ESP valves necessary) is filled with the given **pressure value p_5**. The whole **filling time t_6** is to be investigated by the customer.在加注循环中，对整个制动系统包括液压单元第二回路（ABS/ESP相应阀组需要启动）加注至规定压力值**p_5**。整个加注时间**t_6**由客户进行确认。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Process parameter 过程参数:

Process step:	Test parameter:	Physical values:
6	**Minimum** required absolute **pressure value p_5** at the brake fluid reservoir and at the wheel callipers:储液壶和制动轮缸中要求的最小绝对压力值**p_5**	**≥ y mbar**
	Filling time t_6加注时间**t_6**	**x s**

Notes 注意:

The with x marked **filling time t_6** must be determined by trials with components close to series and the production equipment. These trials have to be executed by the customer.
数值为x的加注时间**t_6**必须由接近生产件水平的部件和生产设备通过试验来确定。这些试验必须由客户来进行。

The with y marked pressure value is to be inferred from the chapter Process data (absolute pressure and time values).
数值为 y 的压力值可由“过程数据（绝对压力和时间值）”相关章节中得出。
The evacuation and filling equipment must be able to reach a sufficient filling pressure (≥ pressure value p_5) in the brake fluid reservoir.
The filling time t6 to reach the defined **pressure value p_5** in the whole brake system is a function of the volumes of the whole brake system, the wheel brake callipers, the pipes and hoses and the size of the orifices of the ABS / ESP hydraulic unit. This time also depends on the capacity of the evacuation and filling equipment. The use of a lower pressure value p_5 means an extension of the filling time.
抽真空加注设备应能在储液壶中实现充分的加注压力（≥压力值**p_5**）。
在整个制动系统中实现规定压力值**p_5**所需的加注时间**t_6**与整个制动系统容积、制动轮缸、制动管路或软管、液压单元截流口孔径都有关系。同时抽真空加注设备的容量对加注时间也有影响。如果压力值**p_5**减小，则加注时间也将相应延长。

The **maximum pressure value p_5 depends** on the burst pressure of the **brake fluid reservoir**. This value must be asked at the manufacturer of the reservoir.
p_5的最大值取决于储液壶的爆裂压力，该值必须向储液壶制造商确认。
During the **fill cycle** the front axle will reach the pressure value p_5 before the rear axle, because of the bigger orifice diameters in the ABS / ESP hydraulic unit.
The time to reach the **pressure value p_5** in the **rear axle** defines the whole **filling time t_6**.
The required filling time t_6 for the production cycle can only be determined by trials with production parts or parts which are close to the series. These trials must be done with the series equipment.
在加注循环中，前轴将先于后轴达到压力值**p_5**，这是因为前轴对应的液压单元截流口孔径更大。
后轴制动轮缸达到压力值**p_5**所需的时间决定了加注时间**t_6**。
生产线上的加注时间**t_6**只能由生产件或接近生产件水平的部件通过试验来确定。这些试验必须通过生产设备来进行。

An **activation** of the **ABS / ESP valves** is necessary **to evacuate the secondary circuits of the hydraulic unit**. Details to the required valve activation see in the chapter Diagnostic step 3: Start of

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

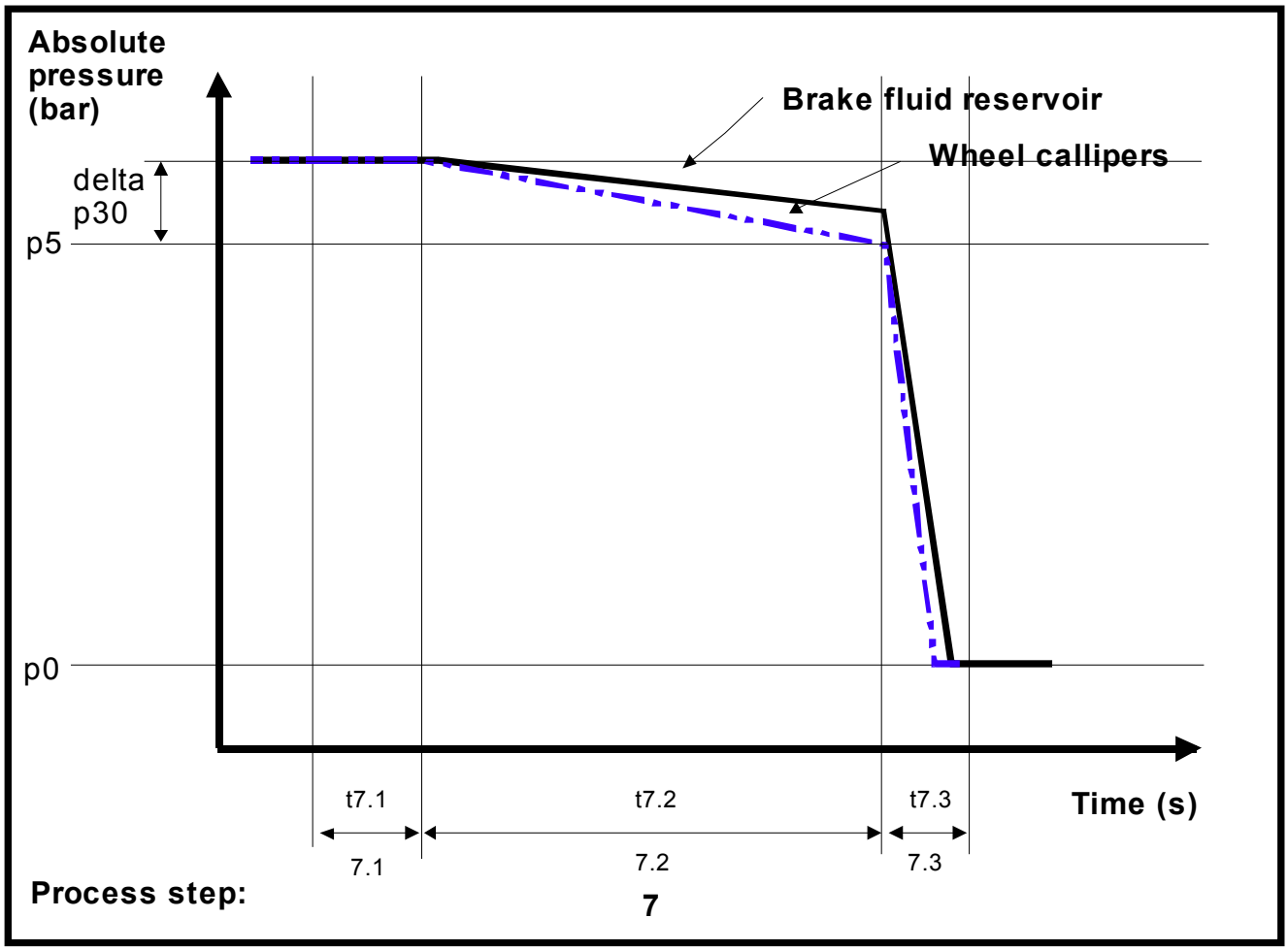
the routine „evacuation and filling“ to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit.
为了排空液压单元第二回路，必须启动 ABS/ESP 相应阀组。具体请参考“诊断步骤 3”相关章节。

Fill cycle ok	加注合格	⇒	**Continue** 继续
Fill cycle nok	加注不合格	⇒	**Abort, print out fault message** 退出，打印错误信息

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

4.7 Process step 7 步骤 7: Pressure test 压力测试

Diagram „Pressure test“:



Details to the diagram „Pressure test“:示意图解析“压力测试”

Process step:	Time:	Process description:
7.1	$t_{7.1}$	**Stabilisation** of the **filling pressure p_5** in the whole brake system. 整个制动系统加注压力**p_5**处于稳定状态。
7.2	$t_{7.2}$	Measurement of the initial and final pressure during the **test time $t_{7.2}$** and evaluation of the occurred **pressure decrease Δp_{30}**.测量**$t_{7.2}$**前后压力，评估压力减小量**Δp_{30}**。
7.3	$t_{7.3}$	**Pressure release** to **ambient-air pressure p_0**. 压力下降至环境大气压**p_0**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Process parameter 过程参数:

Process step:	Test parameter:	Physical values:
7.1	**Filling pressure stabilisation time $t_{7.1}$:**加注压力稳定时间 $t_{7.1}$	**x s**
	Filling pressure value p_5:加注压力值p_5	**≥ y bar**
7.2	Maximal allowed **pressure decay Δp_{30}:**最大允许压力减小值Δp_{30}	**x mbar**
	which is allowed at the brake fluid reservoir for the **test time $t_{7.2}$**::测试时间$t_{7.2}$	**x s**
7.3	**Pressure release time $t_{7.3}$:** 压力下降时间$t_{7.3}$	**x s**
	Atmospheric **ambient-air pressure p_0:** 环境大气压p_0	**~ y bar**

Notes:注意

All with x marked time or pressure values depend on the evac/fill equipment and the maximal available cycle time. Therefore the **total time t_7 (consisting of $t_{7.1}$, $t_{7.2}$ and $t_{7.3}$)** must be determined by the customer.
数值为x的所有时间或压力值取决于抽真空加注设备和最大可用循环时间。因此整体时间t_7（包括$t_{7.1}$, $t_{7.2}$ 和 $t_{7.3}$）应由客户定义。
All with y marked time or pressure values are to be inferred from the chapter Process data (absolute pressure and time values).
数值为 y 的所有时间或压力值可由“过程数据（绝对压力和时间值）”相关章节中得出。

The pressure test was designed to check the tightness of the whole brake system after the filling process. This test doesn't affect the general evacuation and filling process, however, is an additional test step.
压力测试用来在加注过程之后测试整个制动系统的气密性。作为一个额外的测试步骤，该测试不影响整个抽真空加注循环。

The evacuation and filling equipment isolates the pressure pump after the **stabilization time $t_{7.1}$**. Before and after the **test time $t_{7.2}$** it measures the actual pressure values. Afterwards it compares the measured values with given limits (maximally permissible **pressure decay Δp_{30}**). At the end of this test the pressure is released to **air-ambient pressure p_0** within the **test time $t_{7.3}$**.
抽真空加注设备在稳定时间$t_{7.1}$后断开压力泵，然后测量$t_{7.2}$前后的实际压力值，并比较测量值是否在允许范围内（最大允许压力减小值Δp_{30}）。在测试的最后，将压力在$t_{7.3}$内下降到环境大气压p_0。

The ABS respectively ESP valves and the pump motor engine may not be activated during the test!
ABS/ESP 的阀组和泵马达在整个测试期间不能启动。

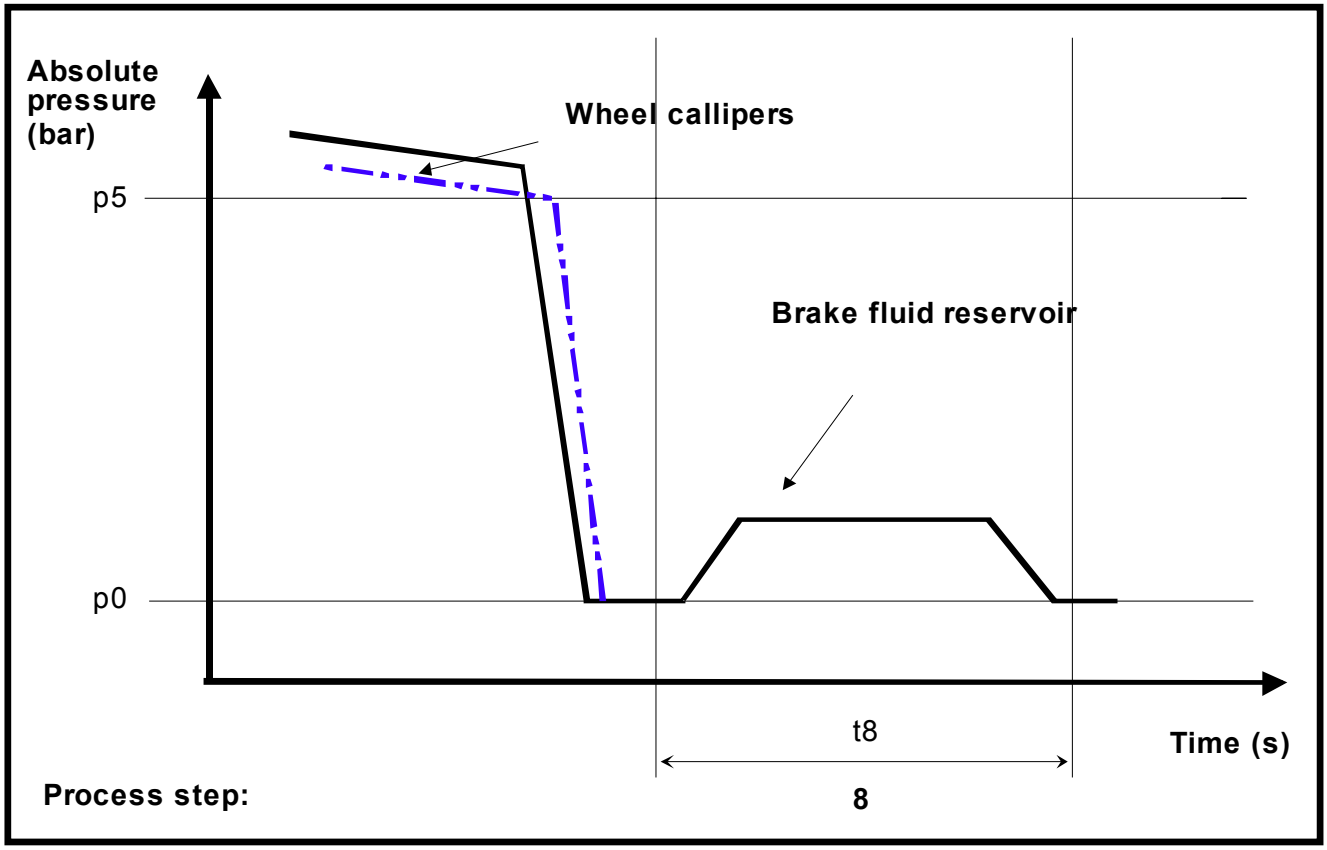
© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Pressure test ok 压力测试合格 ⇒ **Continue** 继续
Pressure test nok 压力测试不合格 ⇒ **Abort, print out fault message**
退出，打印错误信息

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

4.8 Process step 8: Levelling of the brake fluid level in the brake fluid reservoir
步骤 8：加注制动液至储液壶中液面正确位置

Diagram „Levelling of the brake fluid level in the brake fluid reservoir“:



Details to the diagram „Levelling of the brake fluid level in the brake fluid reservoir“:

Process Step:	Time:	Process description:
8	t_8	During the **brake fluid levelling time t_8** is the **brake fluid level in the reservoir** levelled.在加注满制动液时间**t_8**内，制动液液面上升至壶中正确位置。

Process parameter 过程参数:

Process step:	Test parameter:	Physical values:	Conditions:
8	**Brake fluid levelling time t_8:**加注满制动液时间**t_8**	**x s**	**Brake fluid in the reservoir is filled respectively sucked out till the MAX – label is reached** 加注（或抽回）制动液直至壶中液面上限标识处

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Notes 注意:

The with x marked time value depends on the evac/fill equipment and the maximal available cycle time. Therefore the **brake fluid levelling time t_8** must be determined by the customer.
数值为x的时间值取决于抽真空加注设备和最大可用循环时间。因此加注满制动液时间t_8应由客户定义。

The brake fluid in the reservoir must be filled respectively sucked out till the MAX – label is reached.
储液壶中制动液必须加注（或抽回）到液面上限标识处。

The required time for this step must be determined by the customer.
该步骤所需时间应由客户决定。

Brake fluid level ok 制动液液面合格 ⇒ **Continue** 继续
Brake fluid level nok 制动液液面不合格 ⇒ **Abort, print out fault message**
退出，打印错误信息

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

4.9 Process step 9 步骤 9: End of process/handling 2 结束过程/操作 2

End of process / handling 2	
Operator:	**Evacuation and filling equipment/ ECU:**
☞ **Switching off of the ignition (required in case of connection via the OBD2 diagnostic interface)** **Switching off of the external power supply (required in case of direct connection at the ABS or ESP ECU)** 关闭点火开关（适用于通过 **OBD2** 诊断接口连接） 关闭外接电源（适用于通过 **ABS/ESP ECU** 直接连接） ☞ **Disconnect all electrical connections** 断开所有电气连接 ☞ **Disconnect the fill head** 断开加注头	**Switching off of the external power supply (direct connection)** 关闭外接电源（直接连接时）

Process step 步骤:	Parameter 参数:	Physical values 数值:
9	Handling time t_9	x s

Notes 注意:

The with x marked **handling time t_9** depends on the mounting position of the brake fluid reservoir, the hydraulic unit or the position of the diagnosis interface OBD2 in the vehicle interior. The handling time t9 must be determined together with the operators by trials. These trials are executed by the customer.
数值为x的操作时间t_9由储液壶的安装位置、液压单元的位置以及OBD2 诊断接口的位置决定，需要客户通过试验确定。

At first the external power supply or ignition must be switched off. In case of using of an external power supply it must be switched off first, before the operator does disconnect the electrical adaptations.
外接电源或点火开关应当最先关闭。如使用外接电源，操作员应当在断开电气连接前将外接电源关闭。

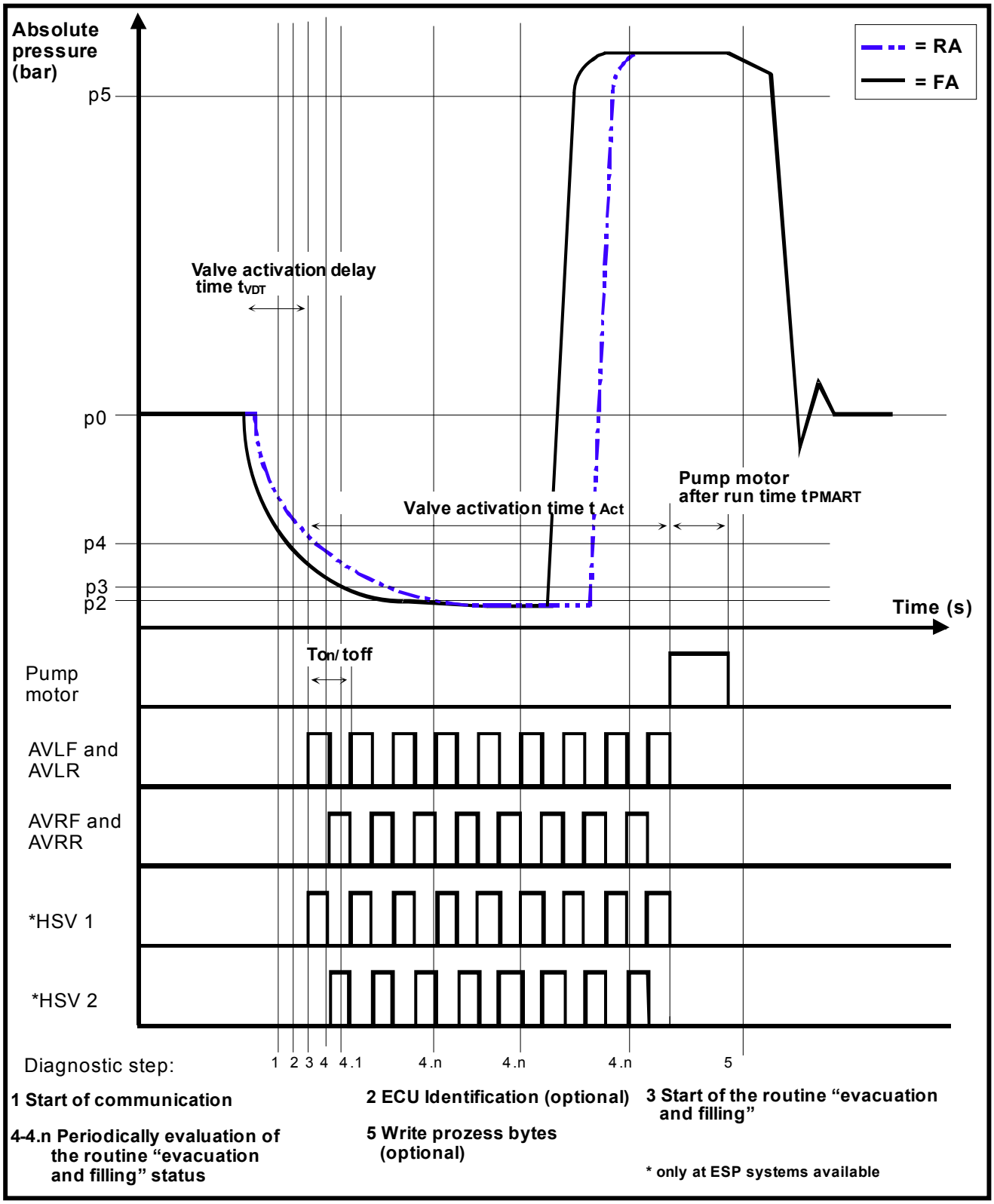
© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

To avoid a reinitialisation of the ABS / ESP ECU, the diagnosis device should not leave the diagnosis with the command "end of the diagnosis" (this starts again the fault detection and storing mechanism). Instead of the power supply should be switched off, otherwise the ABS / ESP ECU will not leave the evacuation and filling process without fault entries.
At different customer's projects a complete avoidance of fault entries is not possible caused by different reasons.
为了避免 ABS/ESP ECU 重启，不能通过“结束诊断”命令来退出诊断服务（该命令将再次启动故障检测和储存机制），而应当通过关闭电源来退出。否则 ABS/ESP ECU 将在退出抽真空加注过程时留下故障记录。
即使这样，要在所有客户的项目中都避免留下故障记录仍然是不可能的，原因有许多种。
More details are to be inferred from the customer diagnosis, evac/fill and repair bleed documentations.
更多详情请参考客户的诊断、抽真空加注和人工排气文件。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

5 Description of each single diagnostic step 每个单独诊断步骤的描述

Diagram about all required diagnostic functions, valve and pump motor activations for generation 9:
9 代系统所有诊断功能和阀/泵马达启动示意图



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Notes:注意

The described diagnostic sequence is an example and based on the diagnosis functions recommended by Bosch (e.g. diagnosis routine "evacuation and filling"). At customer projects where the mentioned diagnosis functions do not exist, the necessary valve and pump motor activations etc. must be identically executed with the existing customer diagnosis commands. To ensure that all required activations were executed from the beginning up to the end of the whole evacuation and filling process correctly, the corresponding status information must be checked by means of the existing customer specific diagnostic functions.
上面描述的诊断序列是基于博世推荐的诊断功能（例如：诊断程序“抽真空加注”）来实现的一个范例。在有些客户项目中，上述的诊断功能并不存在，此时阀/泵马达必须按照同样的程序由现有的客户诊断命令来启动。为了保证所有要求启动的阀/泵马达在整个抽真空加注过程中都运行无误，相对应的状态信息应当通过现有的客户诊断功能来检测。

Important note to the interrogation of the system status (valve relay):
询问系统状态（阀继电器）的注意事项：

With generation 9 the valve relay is always switched off in the normal state. In dependence of the selected diagnostic mode or first with starting of valve activation the valve relay is turned on. Therefore an interrogation of the valve relay to judge the system status is only limited helpful. However this evaluation can be applied to ensure a correct activation nevertheless in the individual or customer specific case, but this important remark must always be considered.
9 代系统的阀继电器在正常状态下总是断开的。选择诊断模式或启动电磁阀时，阀继电器都将接通。因此询问阀继电器状态只能在一定程度上判断出系统状态。尽管这种判断方法只适用于个别情况或指定客户，但是在任何情况下都应予以重视。

Important notes to the control of the valves:
阀组控制的注意事项：

The **valve activation** with the **defined duty cycle t_{on} to t_{off}** is limited to the **maximum valve activation time t_{Actmax}** to avoid an overheating of the valve coils. To guarantee that the maximum valve activation time t_{Actmax} is not exceeded by faulty tester activation, the valve control is automatically reset by the valve protection measures. More detailed information can be inferred from the chapter Valve protection measures at generation 9.
为了避免电磁线圈过热，占空比**t_{on}/t_{off}**下所允许的阀最长启动时间被限制在**t_{Actmax}**。为了保证即使在错误的诊断仪启动命令下阀启动时间也不超过**t_{Actmax}**，在错误命令下阀保护措施将自动重启阀控制。具体请参考“9 代系统的阀保护措施”相关章节。

If the **total process time exceeds the maximal allowed valve activation time t_{Actmax}**, Bosch recommends the introduction of a **valve activation delay time t_{VDT}**.
It is not allowed to repeat a complete valve activation cycle within a short time, because the coils can be damaged by overheating. In case of repetition of complete valve activation cycle a **minimum cooling time $t_{Coolmin}$** must be kept.
如果整个抽真空加注过程时间超过了阀最长允许启动时间**t_{Actmax}**，博世推荐引入阀启动延迟时间**t_{VDT}**。在短时间内重复整个阀启动循环是绝不允许的，因为电磁线圈将因过热而受损。如果需要重复整个阀启动循环，请至少等待一段最短冷却时间**$t_{Coolmin}$**。

Important notes to the pump motor activation 泵马达启动时的注意事项:

An activation of the pump motor (PM) during the whole evacuation process is not allowed, because this can lead to the damage of the pump elements. In addition a pump motor activation is not neces-

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

sary during the whole evacuation and filling process, because a perfect filling result is also reached without a pump motor control.
But at the end of entire evacuation and filling process or during another production step the pump motor must be activated for the defined **pump motor after run time t_{PMART}** to discharge the accumulator chambers. The discharging of the accumulator chambers is absolute required and must be guaranteed, because with filled accumulator chambers no correct ABS control is possible.
在整个抽真空过程中不允许启动泵马达，这是为了避免泵单元干转受损。此外，即使泵马达不启动也可以达到良好的加注效果。
在整个抽真空加注过程的最后阶段或在后续生产步骤中，泵马达必须启动一段规定的后续运行时间**t_{PMART}**以排空蓄能器。如果不排空蓄能器，则ABS无法正常工作。
To avoid damages at the pump elements the maximum **pump dry run time t_{PEdry}** must be considered. It must be guaranteed that the maximum pump dry run time t_{PEdry} is not exceeded during the entire evacuation and filling and a possible repair bleeding process!
For more details refer the chapter Hydraulic data.
为了避免泵单元损坏，必须保证整个抽真空加注过程以及人工排气过程中泵干转时间不超过最长允许干转时间**t_{PEdry}**。具体请参考“液压数据”相关章节。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

5.1 Diagnostic step 1 诊断步骤 1: Start of communication 开始通讯

Start of communication 开始通讯	
Diagnostic unit 诊断仪	**ECU 电子控制单元**
Start of communication 开始通讯	**Change into the customer specific diagnostic mode 进入客户诊断模式**
More information to the chapter "start of communication" is to be inferred from the customer specific documentations.关于"开始通讯"更多信息请参考客户诊断功能描述。	

Notes 注意:

The communication between the tester and ECU can be take place immediately after switching on of the power supply or ignition.
打开电源或点火开关后将立刻开始诊断仪与 ECU 之间的通讯。
In case of a not successful execution of the function „start of communication", it should be repeated minimum 3 times before the process is interrupted.
如果"开始通讯"功能执行不成功，请在中断过程之前至少重复执行 3 次。
More details to the start of communication are to be inferred from the customer specific documentations.关于"开始通讯"更多信息请参考客户诊断功能描述。

Start of communication successful 开始通讯成功 ⇒ **Continue 继续**
Start of communication not successful 开始通讯不成功 ⇒ **Abort, print out fault message 退出，打印错误信息**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

5.2 Diagnostic step 2 诊断步骤 2: ECU identification(optional) 电子控制单元识别（可选）

ECU Identification 电子控制单元识别	
Diagnostic unit 诊断仪	**ECU 电子控制单元**
ECU identification 电子控制单元识别	**Output of the requested ECU data.电子控制单元数据输出**
More information to the chapter "ECU identification" is to be inferred from the customer specific documentations.关于"电子控制单元识别"更多信息请参考客户诊断功能描述。	

Notes 注意:

The to be evaluated ECU identification data must be confirmed by the customer and its assembly plants. This step is an optional step and demands on the requests of the customer.
待评估的电子控制单元识别数据必须由客户及其装配工厂确认。该步骤为可选步骤，取决于客户要求。

The ECU identification data must be compared with the in the tester software stored desired ECU data. This is required to select the right valve activation sequence. The ECU identification data might be changed in case of an ABS/ESP software update respectively modification, therefore it must be easily possible with editor menu to update the data base of the tester respectively evacuation and filling equipment.
电子控制单元识别数据必须和诊断仪中存储的默认数据进行对比，以确保所选择的阀启动程序正确无误。当 ABS/ESP 软件升级时，电子控制单元识别数据将发生变更，因此诊断仪和抽真空加注设备中的数据库应当能用编辑菜单进行方便的升级。
More details to the chapter "ECU identification" are to be inferred from the customer specific documentations. 关于"电子控制单元识别"更多信息请参考客户诊断功能描述。

Known system 已知系统 ⇒ **Continue 继续**
Unknown system 未知系统 ⇒ **Abort, print out fault message 退出，打印错误信息**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

5.3 Diagnostic step 3: Start of the routine „evacuation and filling“ to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit
诊断步骤 3：对 ABS/ESP 液压单元第二回路开始“抽真空加注”程序

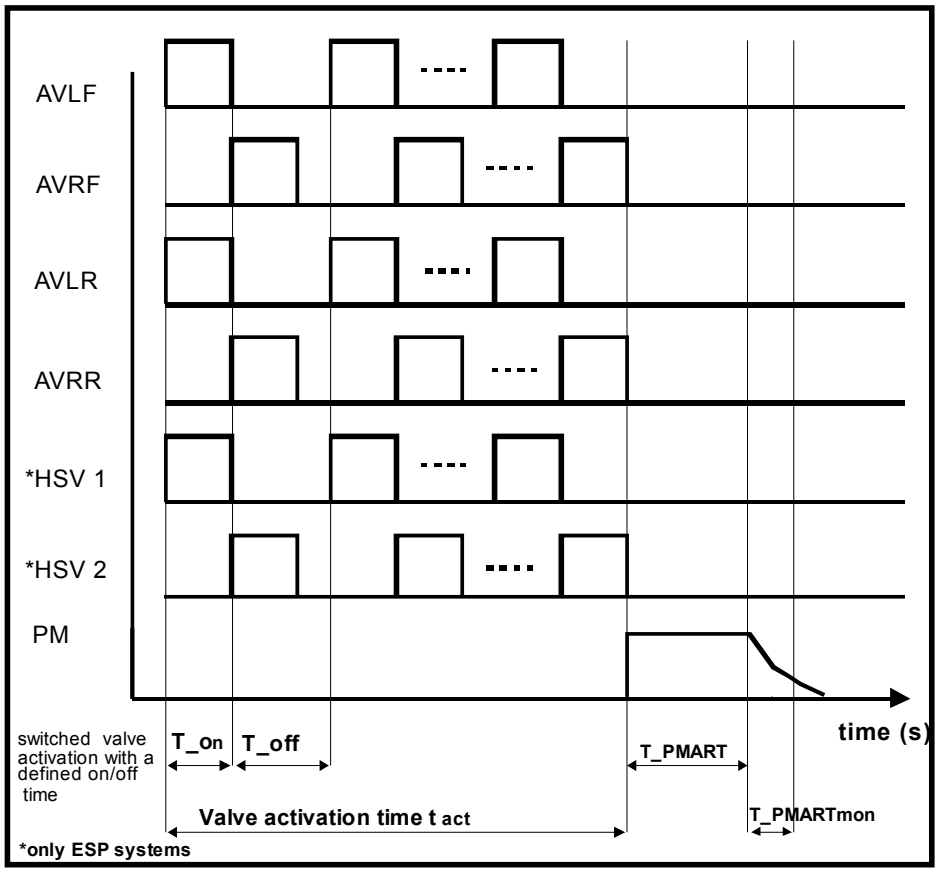
Start of the routine „evacuation and filling“开始“抽真空加注”程序	
Diagnostic unit 诊断仪	**ECU 电子控制单元**
Start of the routine „evacuation and filling“ 开始“抽真空加注”程序	**Activation of the valves and the pump motor to discharge the accumulator chambers (optional) via the routine „ evacuation and filling“** 在“抽真空加注”程序中启动阀/泵马达以排空蓄能器（可选）
More information to the chapter “Start of the routine “evacuation and filling” to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit“ is to be inferred from the customer specific documentations.关于“对 **ABS/ESP** 液压单元第二回路开始抽真空加注程序” 更多信息请参考客户诊断功能描述。	

Notes 注意:

The described sequence is valid for all ABS/ ESP 9 systems. An activation of the corresponding valves via the routine „evacuation and filling“ is required to fill the secondary circuits of a ABS/ESP hydraulic unit.
上述程序适用于所有 ABS/ESP 9 系统。加注 ABS/ESP 液压单元第二回路必须通过“抽真空加注”程序启动相应阀。

The tester activates the corresponding outlet and priming valves (priming valves are only available with ESP systems) with a defined duty cycle **duty cycle on time t_{on} and duty cycle off time t_{off}.** This specific activation profile is realised with the routine “evacuation and filling”:
诊断仪以规定的占空比（高电平时间**t_{on}**，低电平时间**t_{off}**）启动相应出液阀和吸入阀（吸入阀只限于ESP系统）。这里阀启动曲线图是通过“抽真空加注”程序来实现的。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.



The **optional pump motor** activation is only required to discharge the accumulator chambers. This can either be done as described in this chapter, at the end of the filling process, at each following assembly process, at the end of line tests or wherever diagnostics are possible.

The discharging of the accumulator chambers is absolute required and must be guaranteed by the customer, because with filled accumulator chamber an ABS control is only limited possible.

The pump motor must be activated for the specified minimum time „**pump motor after run time t_{PMART}**“.

启动泵马达（可选）仅用于排空蓄能器。该步骤可以在加注过程结束时进行，可以在后续的每一个生产过程中进行，也可以在下线检测时或任何诊断功能可用的情况下进行。

客户必须保证蓄能器的排空，因为在蓄能器未排空情况下 ABS 将无法正常工作。

泵马达启动的最短时间必须达到“泵马达后续运行时间t_{PMART}”。

The required **valve activation time t_{Act}** must be determined by trials executed at the customer. These trials have to be executed with components close to series and the production evac/fill equipment.

Please note that the valves have to be actuated for **the minimum valve activation time t_{Actmin}** to reach the best possible vacuum within the secondary respectively priming circuit of the ABS/ESP hydraulic unit. This is especially valid after execution of an optional compressed air leakage test. The minimum valve activation time t_{Actmin} is the minimal time for the valve activation during the vacuum phase. This time don't include the required valve activation time for the filling phase.

阀启动时间t_{Act}必须由接近生产件水平的部件和在线的抽真空加注设备通过试验来确定。这些试验必须由客户来进行。

请注意，阀启动时间必须至少达到“阀最短启动时间t_{Actmin}”，此时ABS/ESP液压单元第二回路和吸入阀回路才能实现最好的抽真空效果，这一点特别适用于在压缩空气泄漏测试（可选）刚结束之后。阀最短启动时间t_{Actmin}只限于抽真空阶段，并不包括加注阶段所要求的阀启动时间。

<u>Example how to calculate the total valve actuation time t_{Act} of an evacuation and filling process:</u>

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

范例：如何计算抽真空加注过程中的阀启动整体时间t_{Act}

Process step: 步骤	Name of process step: 步骤名称	Process time: 过程耗时	Physical value (s)
3	Evacuation cycle 1 抽真空循环 1	t_3	45
4	Leakage test 泄漏测试	t_4	---
5	Evacuation cycle 2 抽真空循环 2	t_5	5
6	Fill cycle 加注循环	t_6	30
Total	Valve activation time 阀启动时间	t_{Act}	80

Important notes to the control of the valves:
阀组控制的注意事项：

The **valve activation** with the **defined duty cycle t_{on} to t_{off}** is limited to the **maximum valve activation time t_{Actmax}** to avoid an overheating of the valve coils. To guarantee that the maximum valve activation time t_{Actmax} is not exceeded by faulty tester activation, the valve control is automatically reset by the valve protection measures. More detailed information can be inferred from the chapter Valve protection measures at generation 9.
为了避免电磁线圈过热，占空比**t_{on}/t_{off}**下所允许的阀最长启动时间被限制在**t_{Actmax}**。为了保证即使在错误的诊断仪启动命令下阀启动时间也不超过**t_{Actmax}**，在错误命令下阀保护措施将自动重启阀控制。具体请参考“9 代系统的阀保护措施”相关章节。

If the **total process time exceeds the maximal allowed valve activation time t_{Actmax}**, Bosch recommends the introduction of a **valve activation delay time t_{VDT}**.
It is not allowed to repeat a complete valve activation cycle within a short time, because the coils can be damaged by overheating. In case of repetition of complete valve activation cycle a **minimum cooling time $t_{Coolmin}$** must be kept.
如果整个抽真空加注过程时间超过了阀最长允许启动时间**t_{Actmax}**，博世推荐引入阀启动延迟时间**t_{VDT}**。在短时间内重复整个阀启动循环是绝不允许的，因为电磁线圈将因过热而受损。如果需要重复整个阀启动循环，请至少等待一段最短冷却时间**$t_{Coolmin}$**。

More details to the chapter “Start of the routine “evacuation and filling” to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit” are to be inferred from the customer specific documentations.
关于“对 ABS/ESP 液压单元第二回路开始抽真空加注程序” 更多信息请参考客户诊断功能描述。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Diagnostic parameter:诊断参数：

Parameter:	Physical values:
Valve activation delay time t_{VDT}阀启动延迟时间	**x s**
Valve activation time t_{Act}阀启动时间	**x s**
Min. valve activation time t_{Actmin}最短阀启动时间	**y s**
Max. valve activation time t_{Actmax}最长阀启动时间	**y s**
Duty cycle on time t_{on}高电平时间	**y s**
Duty cycle off time t_{off}低电平时间	**y s**
Pump motor after run time t_{PMART} 泵马达后续运行时间	**y s**
Minimum cooling time $t_{Coolmin}$最短冷却时间	**y min**

All with x marked times are project respectively customer specific. Therefore these times must be determined by the customer.
数值为 x 的所有时间值都是随项目或客户不同而变化，因此必须由客户确认。
All with y marked times are to be inferred from the chapter Diagnostic data (times).
数值为 y 的所有时间值可由“诊断数据（时间）”相关章节中得出。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

5.4 Diagnostic step 4: Evaluation of the status of the routine „evacuation and filling“
诊断步骤 4：评估“抽真空加注”程序状态

Evaluation of the status of the routine „evacuation and filling“	
Diagnostic unit 诊断仪	**ECU** 电子控制单元
Evaluation of the status of the routine „evacuation and filling“ 评估“抽真空加注”程序状态	**Output of the status of the routine and the pump motor after run monitoring time $t_{PMARTmon}$** 程序状态与泵马达后续运行监测时间$t_{PMARTmon}$输出
More information to chapter “Evaluation of the status of the routine ‘evacuation and filling’ “ is to be inferred from the customer specific documentations. 关于“评估抽真空加注程序状态”更多信息请参考客户诊断功能描述。	

Notes 注意:

To identify if the routine „evacuation and filling“ inclusive all valve activations are executed without any faults, the actual status of the routine must be **periodically** evaluated. The kind of evaluation and the required diagnostic function is customer specific.
The depending on if the PM should be activated via the routine “evacuation and filling“ the result byte 2 **“pump motor after run monitoring time $t_{PMARTmon}$“** can be **optional** evaluated to ensure that the pump motor was activated to discharge the accumulator chambers.
为了确认“抽真空加注”程序（包括所有阀启动动作）是否被准确无误的执行，必须周期性地对程序的实际状态进行评估。评估类型和所需诊断功能由客户指定。
根据是否要求通过“抽真空加注”程序启动泵马达，可以选择是否对结果字节 2“泵马达后续运行监测时间$t_{PMARTmon}$”进行评估以确保排空蓄能器。
More details to the chapter “Evaluation of the status of the routine ‘evacuation and filling’ “are to be inferred from the customer specific documentations. 关于“评估抽真空加注程序状态”更多信息请参考客户诊断功能描述。

Following responses of the ECU are possible:可能的 ECU 回复如下：

Possibility:	Result byte 1 **“Status of the routine“** 结果字节 1 **“程序状态”**	Evaluation (necessary): 评估（非可选）	Result byte 2 **“pump motor after run monitoring time $t_{PMARTmon}$“** 结果字节 2 **“泵马达后续运行监测时间$t_{PMARTmon}$”**	Evaluation (optional): 评估（可选）
1	Valve activation routine is still running 阀启动程序执行中	Correct, but must be repeat 正确但必须重复	Default value 默认值	Not required 不需要

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

2	Valve activation routine successful finished 阀启动程序成功完成	Correct 正确	y	Seconds	Required and correct, if $t_{PMARTmon}$ > $t_{PMARTmon_min}$ else faulty 需要： 如果$t_{PMARTmon}$ > $t_{PMARTmon_min}$ 正确；否则错误
3	Valve activation routine interrupted 阀启动程序中断	Faulty 错误	Default value 默认值		Not required 不需要
4	Valve activation routine not successful finished 阀启动程序完成但不成功	Faulty 错误	Default value 默认值		Not required 不需要

Diagnostic parameter:诊断参数

Parameter:	Physical value:
Min. pump motor after run monitoring time **t_{PMART_min}泵马达后续运行监测最短时间t_{PMART_min}**	**y s**

All with y marked times are to be inferred from the chapter Diagnostic data (times).
数值为 y 的所有时间值可由“诊断数据（时间）”相关章节中得出。

Routine status correct 程序状态正确 ⇒ **Continue 继续**

Routine status during the evacuation cycle not correct
抽真空循环中程序状态不正确 ⇒ **Print out fault message, Abort 退出，打印错误信息**

Routine status during the filling cycle not correct
加注循环中程序状态不正确 ⇒ **Print out fault message, Continue**
继续，打印错误信息

Result byte 2 not correct (optional)
结果字节 2(可选)不正确 ⇒ **Print out fault message, Continue**
继续，打印错误信息

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

5.5 Diagnostic step 5 诊断步骤 5: Write process bytes (optional)写入过程字节（可选）

Write process bytes	
Diagnostic unit 诊断仪	**ECU 电子控制单元**
Write process bytes 写入过程字节	**Writes the process results into the EEPROM of the ABS/ESP system 将抽真空加注过程结果写入 ABS/ESP 系统的 EEPROM**
More information to the chapter "Write process bytes" is to be inferred from the customer specific documentations. 关于"写入过程字节"更多信息请参考客户诊断功能描述。	

Notes 注意:

All with **tbd** (to be defined) marked values are customer specific. **tbd** 值请参考具体客户规范.

With this optional function it is possible to write additional information about the evacuation and filling process into the EEPROM of the ABS/ESP system.
The ABS/ESP ECU is delivered with predefined value (customer specific). This value is written into the ABS/ESP EEPROM by the Bosch plant. In case of none successful communication with the ABS/ESP ECU at the customer assembly plant, the delivery value is still stored. These conditions must be considered as a not successful respectively faulty evacuation and filling process.
The offered EEPROM location to store this production information amounts up to two data bytes (customer specific). The definition and to be written process data is in responsibility of the customer. To do the right evaluation all defined values and data must be hand out to the corresponding suppliers (e.g. end of line equipment).
该可选功能可以在 ABS/ESP 系统的 EEPROM 中写入抽真空加注过程的附加信息。
ABS/ESP ECU 在发送至客户时已由博世工厂在 ABS/ESP EEPROM 中预先写入了一个预定义值（由客户指定）。如果 ABS/ESP ECU 在客户装配工厂中通讯不成功，则该预定义值仍将存储在 EEPROM 中。这类情况必须被视为抽真空加注过程不成功或发生错误。
EEPROM 中有两个字节的地址位用于存储该附加信息（由客户指定）。客户负责过程字节的定义和写入。为了保证后续过程中能对过程字节进行正确评估，所有定义的数据都必须发送给相关联的设备供应商（如下线检测设备）。

Example of writing two process bytes:写入两位过程字节范例：

Evac/fill equipment writes the process bytes with detailed information:抽真空加注设备写入过程字节（带详细信息）			
Fault description of the process byte 1: 过程字节 1 的错误描述	**Fault description of the process byte 2: 过程字节 2 的错误描述**	**PB 1 data to be written (hex):**	**PB 2 data to be written (hex):**
Fault during the evacuation cycle 抽真空循环中发生错误	**Evac/fill facility 1 抽真空/加注设备 1**	**10**	**01**
Entire evacuation and filling process ok 整个抽真空加注过程合格 etc.	**Evac/fill facility 3 抽真空/加注设备 3 etc.**	**tbd etc.**	**03 etc.**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Evac/fill equipment writes the process bytes without detailed information: 抽真空加注设备写入过程字节（无详细信息）			
Fault description of the process byte 1: 过程字节 1 的错误描述	**Fault description of the process byte 2:** 过程字节 2 的错误描述	**PB 1 data to be written (hex):**	**PB 2 data to be written (hex):**
Process ok 过程合格 **Process nok** 过程不合格	**Default respectively delivery value** 默认或预定义值	**tbd** **tbd**	**tbd** **tbd**

Write process bytes ok 写入字节 **ok** ⇒ **Continue** 继续
Write process bytes nok. 写入字节 **nok** ⇒ **Print out fault message, abort** 打印错误信息，退出

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

6 Data 数据

6.1 Process data (absolute pressure and time values) 过程数据（绝对压力和时间值）

Process step:			Parameter:	Condi-tion:	Physical value:	Unit:	Typical val-ues:
1	Handling 1 操作 1		Handling time t_1	~	tbd	Seconds	
2	.1	Compressed air leakage test 压缩空气泄漏测试	Pressure increase time $t_{2.1}$	~	tbd	Seconds	
			Max. pressure p_1	≤	21	bar	6bar
	.2		Max. pressure decay Δp_{10}	≤	tbd	mbar	
			Test time $t_{2.2}$	≤	tbd	Seconds	
	.3		Pressure decrease time $t_{2.3}$	~	tbd	Seconds	
			Ambient-air pressure p_0	≤	1	bar	
3	Evacuation cycle 1 抽真空循环 1		Max. pressure p_2	≤	2	mbar	
			Max. pressure p_3	≤	10	mbar	
			Evacuation time t_3	~	tbd	Seconds	30 to 60s
	.1	Coarse leakage test 大泄漏测试	Max. pressure p_4	≤	200	mbar	
			Delay time $t_{3.1}$	~	tbd	Seconds	
4	Leakage test 泄漏测试		Max. pressure decay Δp_{20}	≤	tbd	mbar	Δ1 mbar
			Test time t_4	≤	tbd	Seconds	5 to 10s
5	Evacuation cycle 2 抽真空循环 2		Evacuation time t_5	~	tbd	Seconds	5s
6	Fill cycle 加注循环		Min. pressure p_5	≥	4	bar	
			Filling time t_6	~	tbd	Seconds	FA: 10 to 25 s RA: 15 to 45 s
7	.1	Pressure test 压力测试	Fill pressure stabilisation time $t_{7.1}$	~	tbd	Seconds	
			Min. pressure p_5	see process step 6			
	.2		Max. pressure decay Δp_{30}	≤	tbd	mbar	
			Test time $t_{7.2}$	≤	tbd	Seconds	

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

	.3		Pressure decrease time $t_{7.3}$	~	tbd	Seconds	
			Ambient-air pressure p_0	see process step 2.3			
8	Levelling of the brake fluid level in the brake fluid reservoir 加注制动液至储液壶中液面正确位置		Brake fluid levelling time t_8	~	tbd	Seconds	
9	Handling 2 操作 2		Handling time t_9	~	tbd	Seconds	

Notes:注意

All with **tbd** (to be defined) marked values are customer specific.**tbd** 数值请参考具体客户规范。
All brake components must be try and able to fulfil the requirements to reach the minimal specified absolute pressure values within the defined times.
所有制动元件必须能够在规定时间内达到指定的最小绝对压力值。
Not degassed brake fluid absorbs at **room temperature** $_{RT}$ and **ambient-air pressure** $\mathbf{p_0}$ up to

5% air

未除去气体的制动液在室温和环境大气压$\mathbf{p_0}$最多将吸收 5%的空气。
The given percentaged value for the amount of air at a not degassed brake fluid depends on the used kind of brake fluid (value is valid for DOT 4).
To reach the best results at a vacuum filling process the brake fluid should be degassed that means it should **not** include more than

10% residual air

of the at room temperature $_{RT}$ and ambient-air pressure p_0 solved air.
未除去气体的制动液中的空气含量百分比取决于所用制动液的种类（该值适用于 DOT4）。
为了达到最好的抽真空加注效果，制动液必须除去气体。这意味着制动液中残留空气含量不能超过室温和环境大气压$\mathbf{p_0}$下其溶解空气量的 10%。

If a degassing or pre-treatment process is not possible, the residual air in the brake system respectively hydraulic unit is not completely dissolved. In case of a short time between the evac/fill and end of line process some additional pressure increase pulses are required to force the air solving process. In case of reaching of a lower absolute pressure value p_3 as specified, the residual air contents of a pre-treated brake fluid can be increased (for more details please see the table below):
如果没有除去气体或预处理过程，制动系统和液压单元中的残留空气将不会完全溶解。如果抽真空加注和下线检测过程之间有时间间隔，这段时间应当用来施加额外的增压脉冲以促使空气溶解在制动液中。如果规定要达到的绝对压力值$\mathbf{p_3}$更小，则预处理制动液中的残留空气含量将增加（具体请参考下表）。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Volume of the brake system in ccm 制动系统容积	Reached absolute **pressure p_3** at the wheel callipers in mbar制动轮缸中绝对压力值**p_3**	**Residual air contents in the brake system** after the evacuation cycle in ccm (%)抽真空循环后制动系统中的残留空气含量	**Residual air contents** in **degassed brake fluid** in comparison to **soaked brake fluid** in ccm (%)除去气体的制动液与饱和制动液中残留空气含量对比	**Total residual air contents after the filling process** in ccm (%)加注过程后残留空气总含量
700	10	7 (1)	3,5 (10)	10,5 (1,5)
	2	1,4 (0,2)	8,8 (25)	10,2 (1,46)

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

6.2 Diagnostic data (times) 诊断数据（时间）

Diagnostic step:		Parameter:	Condi-tions:	Physical value:	Unit:
3	Start of the routine „evacuation and filling“ to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit 对 ABS/ESP 液压单元第二回路开始“抽真空加注”程序	Valve activation delay time t_{VDT}	~	tbd	Seconds
		Duty cycle on time t_{on}	~	0,9	Seconds
		Duty cycle off time t_{off}	~	1,1	Seconds
		Valve activation time t_{Act}	=	tbd	Seconds
		Min. valve activation time t_{Actmin}	≥	20	Seconds
		Max. valve activation time t_{Actmax}	≤	150	Seconds
		Pump motor after run time t_{PMART}	=	2	Seconds
		Min. cooling time t_{Cool_min} (optional)	≥	5	Minutes
4	Evaluation of the status of the routine „evacuation and filling“ 评估“抽真空加注”程序状态	Min. pump motor after run monitoring time $t_{PMART_mon_min}$	=	0,1	Seconds

Notes 注意:

All with **tbd** (to be defined) marked values are customer specific.
tbd 值请参考具体客户规范。
The **maximal valve activation time t_{Actmax}** depends on the activation ratio of the **duty cycle on time t_{on}** to the **duty cycle off time t_{off}**.
阀最长启动时间**t_{Actmax}**取决于占空比高电平时间**t_{on}**与占空比低电平时间**t_{off}**的比值。
More detailed information can be inferred from the chapter Valve protection measures at generation 9.
更多详情请参考“9 代系统阀保护措施”相关章节。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

6.3 Hydraulic data 液压数据

Hydraulic data 液压数据:	Unit:	ABS 9CB	ABS 9CE	ESP 9CB/CE	ESP 9CP
Volume of the primary circuit per circuit (per hydraulic unit):第一回路每条油路容积	cm^3	3,5 (7,0)	3,6 (7,2)	6,1 (12,2)	6,5 (13,0)
Volume of the secondary circuit per circuit (per hydraulic unit):第二回路每条油路容积		0,8 (1,6)	1,2 (2,4)	0,7 (1,4)	0,8 (1,6)
Volume of the priming circuit per circuit (per hydraulic unit):吸入阀回路每条油路容积		---	---	0,8 (1,6)	0,8 (1,6)
Dead space of the pump elements per circuit (per hydraulic unit):泵单元每条油路盲区		0,2 (0,4)	0,2 (0,4)	0,2 (0,4)	0,2 (0,4)
Inlet valve orifice diameter RA:后轴进液阀孔径	mm	customer specific 客户规范			
Inlet valve orifice diameter FA:前轴进液阀孔径		customer specific 客户规范			
Outlet valve orifice diameter RA: 后轴出液阀孔径		customer specific 客户规范			
Outlet valve orifice diameter FA: 前轴出液阀孔径		customer specific 客户规范			
Maximal Pump element dry run time t_{PEdry} :泵单元最长干转时间t_{PEdry}	s	≤ 60			

Notes 注意:

All mentioned volume data are rounded values.
所有上述容积数据都是圆整值。

The evacuation and filling time depends on the used **inlet valve orifice diameter**. The used orifices are project and customer specific.
抽真空加注的时间取决于进液阀孔径，孔径大小请参考具体项目/客户规范。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

6.4 Electrical data 电气数据

6.4.1 Information to the calculated current consumption of an ABS/ESP system ABS/ESP 系统的电流消耗计算值

Calculated current consumption at a 12V voltage supply 12V 电压电源下的电流消耗值				
Component:		**Physical value:**	**Unit:**	**Remarks:**
ECU:		**≤ 0,5**	**A**	
AV coil (7,3 Ω): 出液阀线圈		**1,64**	**A**	
HSV coil (3,6 Ω): 高压开关阀线圈		**3,3/0,97**	**A**	**Valve protection measurement active! For more details please see chapter 7.** 阀保护措施启动！具体请参考第 7 章。
Max. total valve current consumption: 阀组电流消耗最大值	**ABS:**	**3,28**	**A**	**Only valid if the defined valve activation ratio and profile is kept. Details see chapter Diagnostic step 3: Start of the routine „evacuation and filling“ to evacuate and fill the secondary circuits of the ABS/ESP hydraulic unit.** 仅在按照规定的阀启动占空比和曲线图运行时有效。具体请参考“诊断步骤 3”相关章节。
	ESP:	**6,58**		
PM start up current I_{on} 泵马达启动电流		**≤ 145**	**A**	**Depending on the used pump motor.** 由所用泵马达型号决定
PM start up current time t_{on} 泵马达启动电流持续时间		**≤ 20**	**ms**	
PM constant current I_{on_const}泵马达稳定电流		**≤ 15**	**A**	

6.4.2 Specification of the voltage supply 电压规格

Specification voltage supply		
Parameter:	**Physical value:**	**Remarks:**
Range of voltage [V]: 电压范围	**11,5 ≤ power supply unit ≤ 14**	**Allowed voltage range measured at the ECU. It must be guaranteed that the voltage never drops below 10V (e.g. buffering of the voltage with a condenser required)!** 电压允许范围在 **ECU** 处测得。电压必须保证不低于 **10V**（例如使用电容器作为电压缓冲器）
Current [A]:电流	**≥ 100**	**With pump motor activation.**泵马达启动

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

	≥ 20	Without pump motor activation.泵马达不启动

Notes 注意:

The evacuation and filling equipment respectively its power supply hast to meet the above described requirements.
抽真空加注设备及其电源供应必须满足上述要求。
Independent of the length of the wires between the power supply and the ABS/ESP ECU and the possible caused voltage drop the specified electrical values must be kept.
电源与 ABS/ESP ECU 之间线束长度可能引起一定的电压降，但对以上规定的电气数据不构成影响。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

6.4.3 Specification of the ECU connection 电子控制单元连接规范

Specification ECU connector 电子控制单元连接规范		
Parameter:	**Physical value:**	**Remarks:**
Contact resistance R_{cont_resis} [mΩ]:触点电阻	**≤ 10**	**The used ECU connector should be an industrial plug (no series ECU connector) to ensure a safe repeated plugging. Before the electric contacts are closed, the ECU connector must be led in the guidance grooves of the ECU to avoid a pre-damage of the pins. Details are contained in the appendix. The original drawing can be requested at Bosch.ECU** 连接器必须是工业型接头（不是量产型 **ECU** 连接器），这是为了保证可以安全的重复插拔。在闭合电气连接时前，**ECU** 连接器必须沿着 **ECU** 导引槽小心插入避免损伤针脚。具体请参考附录。原始图纸可向博世索取。 **The pins should be designed either as spring loaded (e.g. company Ingun) or flat sockets (e.g. company Odu). This is required to avoid a damage of the ECU pins and to guarantee a safe connection. A connection on the complete contact surface (not only on the top of the contact) is recommended.**针脚应当设计成弹簧负载型或平插座型。这是为了避免损伤 **ECU** 针脚并确保连接安全。推荐使用包容式接触连接（而不仅是顶面接触连接）。 **The industrial connector design must be adapted to the respective ECU plug and the mounting position of the hydraulic unit in the vehicle.**工业型连接器设计必须考虑到液压单元的 **ECU** 插头形状和车内安装位置。
Contact separation time $t_{contact_sep}$ [µs]:触点分离时间	**≤ 10**	
Contact gap (s1) between ground and power terminals [mm]: 接地端与电源端触点间隙	**≥ 2**	

Specification of a 38 pin connector38 针连接器的规范 (ABS and ESP)	
Pin appellation 针脚名称:	针脚号 **Pin- no.:**
Power supply pump motor 泵马达电源线	**1**
Power supply valves 阀电源线	**25**
Power supply ECU 电控单元电源线	**7**
Power supply WAU 唤醒功能电源线	**35**
Ground ECU and valves 阀和电控单元地线	**38**
Ground pump motor 泵马达地线	**13**
Diagnostic connection 诊断连接	**Customer specific** 客户规范

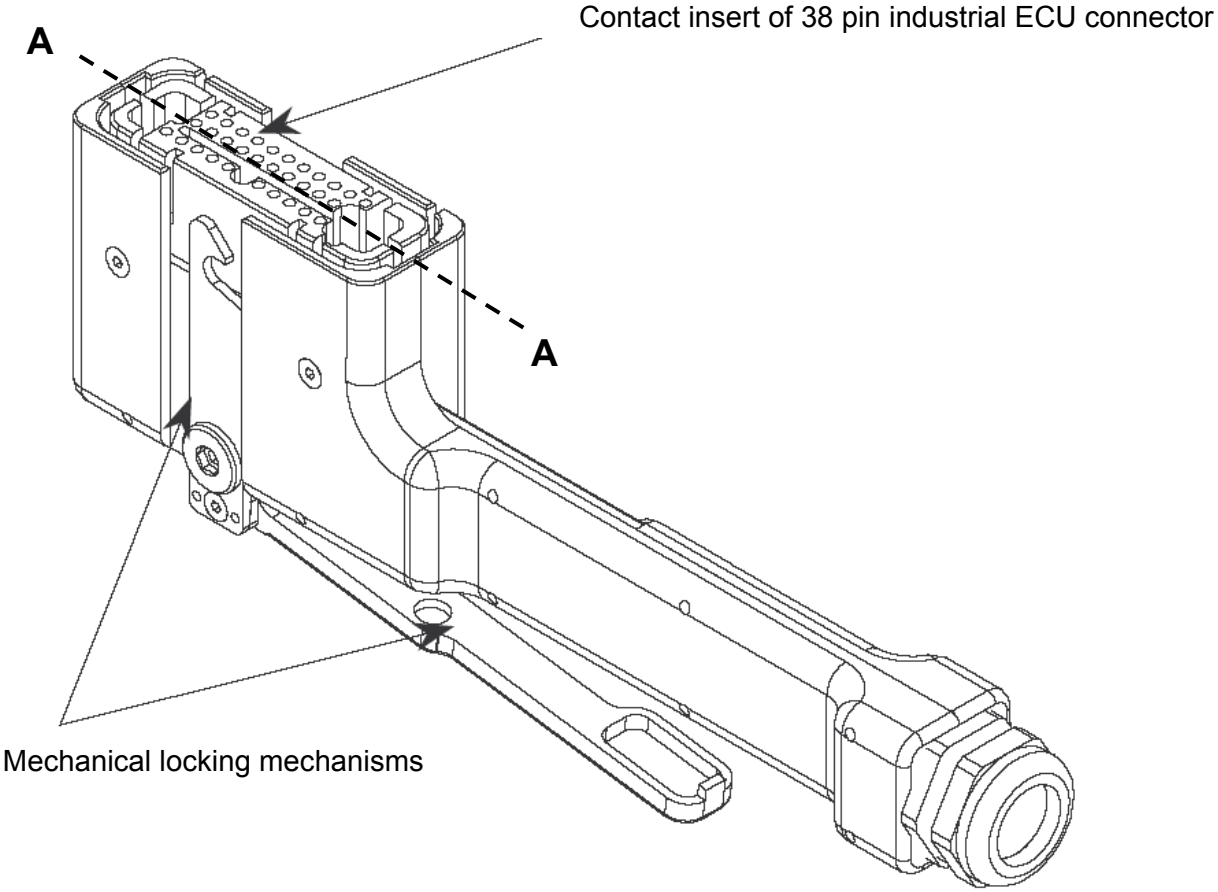
Notes 注意**:**

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

To avoid damages at the ECU caused by an electrical overstress (EOS) following rules have to be considered 为了避免电冲击损坏 ECU 须注意以下几点:

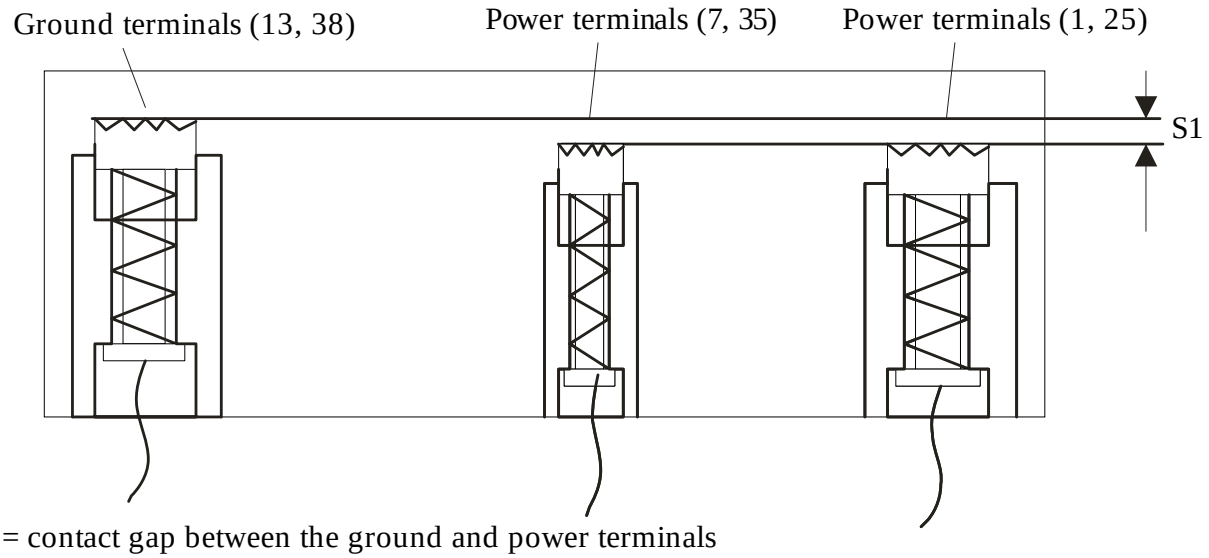
- **It is never allowed to disconnect the ECU connector while a valve or pump motor actuation is running! 阀或泵马达启动时不能拆除 ECU 连接器。**
- **The external power supply has to be completely switched off before the ECU connector is connected respectively disconnected!连接或拆除 ECU 连接器时必须关闭外接电源！**
- **During the electrical connection process the ground terminals of the ECU connector have to contact the ground pins before the power pins within the ECU. This can be guaranteed if there is a contact gap (s1) between the ground and the power terminals within the industrial ECU connector. For more details please refer the sectional drawing (A-A) below.在电气连接过程中，ECU 连接器的接地端应当先接触 ECU 的地线针脚，而后连接器的电源端才接触 ECU 的电源针脚。为了保证这一点，在工业型 ECU 连接器的接地端与电源端之间设计了一个触点间隙（s1）。更多详情请参考下列 A-A 断面图。**

<u>Example of a 38 pin industrial ECU connector:</u>



<u>Sectional drawing (A-A) of a 38 pin industrial ECU connector:工业型 ECU 连接器 A-A 断面图</u>

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.



S1 = contact gap between the ground and power terminals

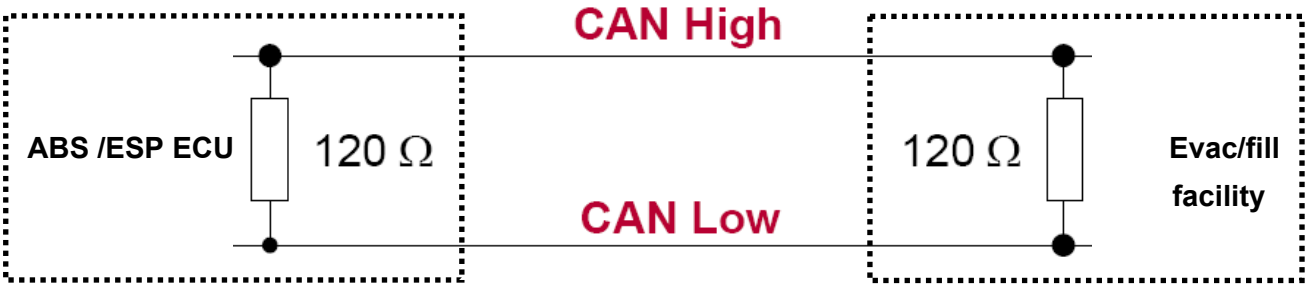
6.4.4 Note „Installation of a termination resistance within the CAN Bus“注意“CAN 终端电阻的安装”。

Precondition for a good CAN signal level (dominant/recessive) respectively a interference-free CAN communication is a correct defined size of the entire CAN termination resistance (more details can take out of the standard ISO 11898 „High-Speed CAN“). Following notes regarding the installation of an additional CAN termination resistance have to be paid attention:

为了得到更好的 CAN 信号并消除 CAN 通讯中的干扰，需要加入正确规格的 CAN 终端电阻。详情请参考 ISO 11898“高速 CAN”。请注意下列有关附加 CAN 终端电阻的安装要点。

Case 1 „direct connection at the **ABS respectively ESP ECU with a CAN termination resistance**”: 在 **ABS/ESP ECU** 直接连接中采用 **CAN** 终端电阻

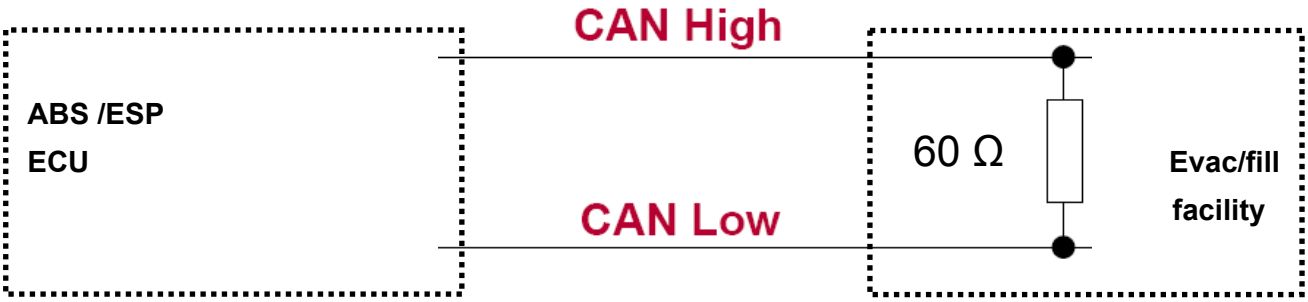
CAN termination resistance CAN 终端电阻	**Ohm**
ABS respectively ESP ECU	120
Evac/fill facility (CAN communication module, electric control box or cable) 抽真空加注设备（CAN 通讯模块、电气控制盒或线缆）	120
Total	**60 (parallel)**



Case 2 „direct connection at the **ABS respectively ESP ECU without a CAN termination resistance**”:在 **ABS/ESP ECU** 直接连接中不采用终端电阻

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

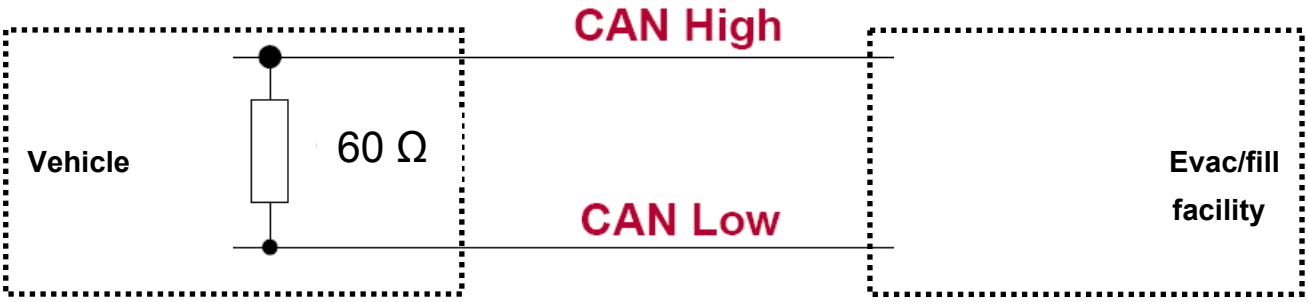
CAN termination resistance CAN 终端电阻	**Ohm**
ABS respectively ESP ECU	0
Evac/fill facility (CAN communication module, electric control box or cable) 抽真空加注设备（CAN 通讯模块、电气控制盒或线缆）	60
Total	**60**



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Case 3 „Connection at the vehicle diagnostic interface (OBD2)”在车辆诊断接口处连接:

CAN termination resistance	Ohm
ABS respectively ESP ECU	60
Evac/fill facility (CAN communication module, electric control box or cable) 抽真空加注设备（CAN 通讯模块、电气控制盒或线缆）	0
Total	**60**



Remark 备注:

Depending on the customer specifics, the CAN termination resistances can be included in different ECU's. Therefore the installation of an additional CAN termination resistance is depending on the used kind of electrical connection and if the ABS/ESP ECU includes a termination resistance or not.

CAN 终端电阻可以安装在各种 ECU 中，具体取决于客户规范。因此附加的 CAN 终端电阻的安装取决于所采用的电气连接以及 ABS/ESP ECU 是否包含终端电阻。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

7 Valve protection measures at generation 9 9 代系统的阀保护措施

	ABS	ESP
Valve:	**Diagnostic mode 诊断模式**	
EV 7,3 Ω	**Not activated 未启动**	
AV 7,3 Ω	**If valve activation value > 0% (via customer diagnosis only full time =100% activation possible); than start the up/down counter and count with a ratio up/down = 2/1; limited to 30s; for each AV valve a separate counter is used** 如果阀启动值 > 0%（在客户诊断中只允许 100%全时启动），则上/下计数器以 2/1 的上/下比开始计数，限时 30s；每个出液阀都有单独的计数器。	
HSV 3,3 Ω	**---**	**20ms with 100% duty cycle, than lowering of the duty cycle to $U_{effective}$ = 3,5V and cyclic all 200ms for 20ms with 100%**以 100%的占空比启动 20ms，然后降低占空比至$U_{effective}$ = 3,5V；每 200ms中 20ms为 100%占空比，循环进行。
USV 7,3 Ω	**---**	**Not activated 未启动**
USV 4,3 Ω		

Notes 注意:

All valve protection measurements are active in the diagnostic mode. These measurements can not be switched off.在诊断模式下，所有阀保护措施都被激活，无法关闭。

Following values for the on-duration of the AV valves in relation to the valve activation ratio of the duty cycle on time t_{on} to the duty cycle off time t_{off} are valid:

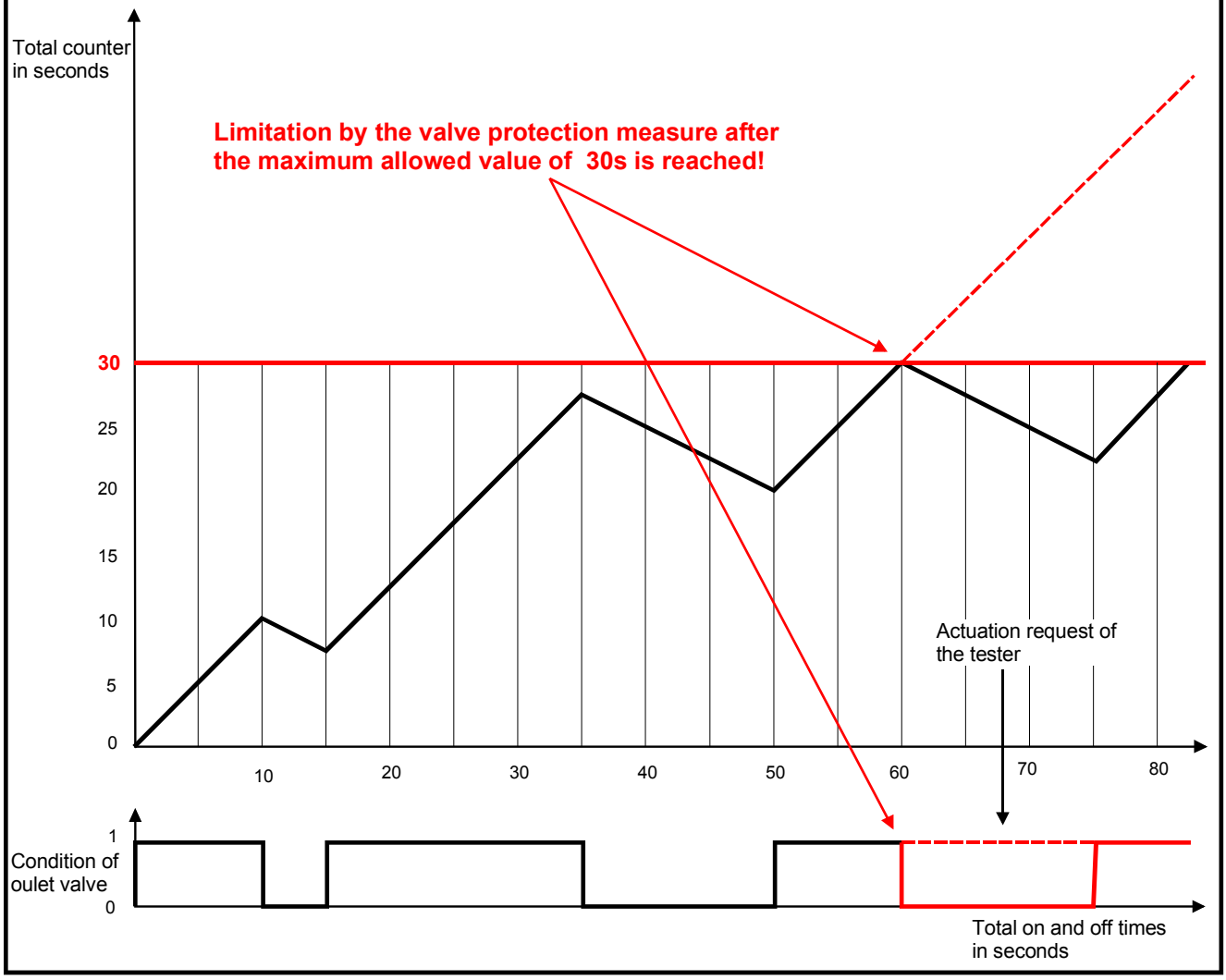
On-duration (%)	Duty cycle on time t_{on} (s)	Duty cycle off time t_{off} (s)	Possible valve activation time t_{Act} (s)
75	1,5	0,50	47,6
50	1	1	117
45	0,9	1,1	168,6

The **maximal allowed valve activation time t_{Actmax}** is to be inferred from the chapter Diagnostic data (times).阀启动最长允许时间t_{Actmax}可由“诊断数据（时间）”相关章节得出。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

Gen 9: Valve protection measure: Up/down counter with a ratio of = 2/1

Example: Outlet valve with a maximum limitation to 30 seconds



Notes 注意:

A forced limitation by the valve protection measurement is active as long as the valve coils and the ECU are protected against overheating. First after a corresponding cooling time the valves can be switched on again.

To ensure the important coil respectively ECU cooling time after a executed valve actuation via diagnosis in a previous ignition cycle (after a hardware or software reset), the actual coil resistance values are measured and used after each power up to initialise the start up value of the up/down counters. Caused by different reasons the described valve protection measurements cannot protect the valve coils and the ECU with 100% safeness. That's the reason why the customer is responsible to ensure a correct valve activation sequence!

阀保护措施强制保护阀电磁线圈与 ECU 免于过热。当冷却时间足够后阀组方能再度启动。有时前一个点火循环中利用诊断功能启动了阀，而再次点火后硬件和软件都重启了。为了保证阀电磁线圈与 ECU 有足够冷却时间，可以通过测量阀线圈的电阻值来判断上/下计数器的合理初始值。出于多种原因，即便是阀保护措施也无法保证阀电磁线圈和 ECU100%安全。因此，客户有责任确保阀启动序列正确！

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

8 Fault handling 错误处理

Fault handling 错误处理:	
Kind of fault 错误类型:	**Fault message 错误信息, fault handling 错误处理:**
Evacuation and filling equipment 抽真空加注设备 **Diagnostic unit 诊断仪**	**Fault message, vehicle marked with a fault message → repair bleed 人工排气** **Fault message, vehicle marked with a fault message → repair bleed 人工排气**

Notes 注意:

In case of occurrence of any electrical or diagnostic failure during **the filling cycle** the standard filling process can be continued. However the evacuation and filling equipment must output respectively print out a fault message to ensure that this vehicle is repair bleed in rework area. In addition optional the process byte can be written with corresponding fault information.
如果加注循环中出现任何电气或诊断错误，标准的加注过程可以继续，但设备必须打印错误信息以确保车辆被送到返工区域进行人工排气。此外，在过程字节中可以写入相应的错误信息（可选）。

In case of occurrence of any failure during **the evacuation cycle** the standard evacuation and filling process should be interrupted, because after the fixing of the problem this vehicle can be filled with a standard evacuation and filling process again.
如果抽真空循环中出现错误，标准的抽真空加注过程将被中断，这是因为问题解决后车辆可以从头开始抽真空加注。

The maximal allowed valve activation times must always be noted!
必须注意阀启动最长允许时间！

To avoid any unnoted pre-damages at the ABS/ESP system caused by a faulty process (e.g. to many valve activation cycles which might caused a overheating of the coils or ECU), the hydraulic unit inclusive ECU should be changed.
为了避免错误过程造成的 ABS/ESP 系统隐性损坏（如过多的阀启动循环将导致阀线圈或 ECU 过热），必须更换液压单元（包括 ECU）。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

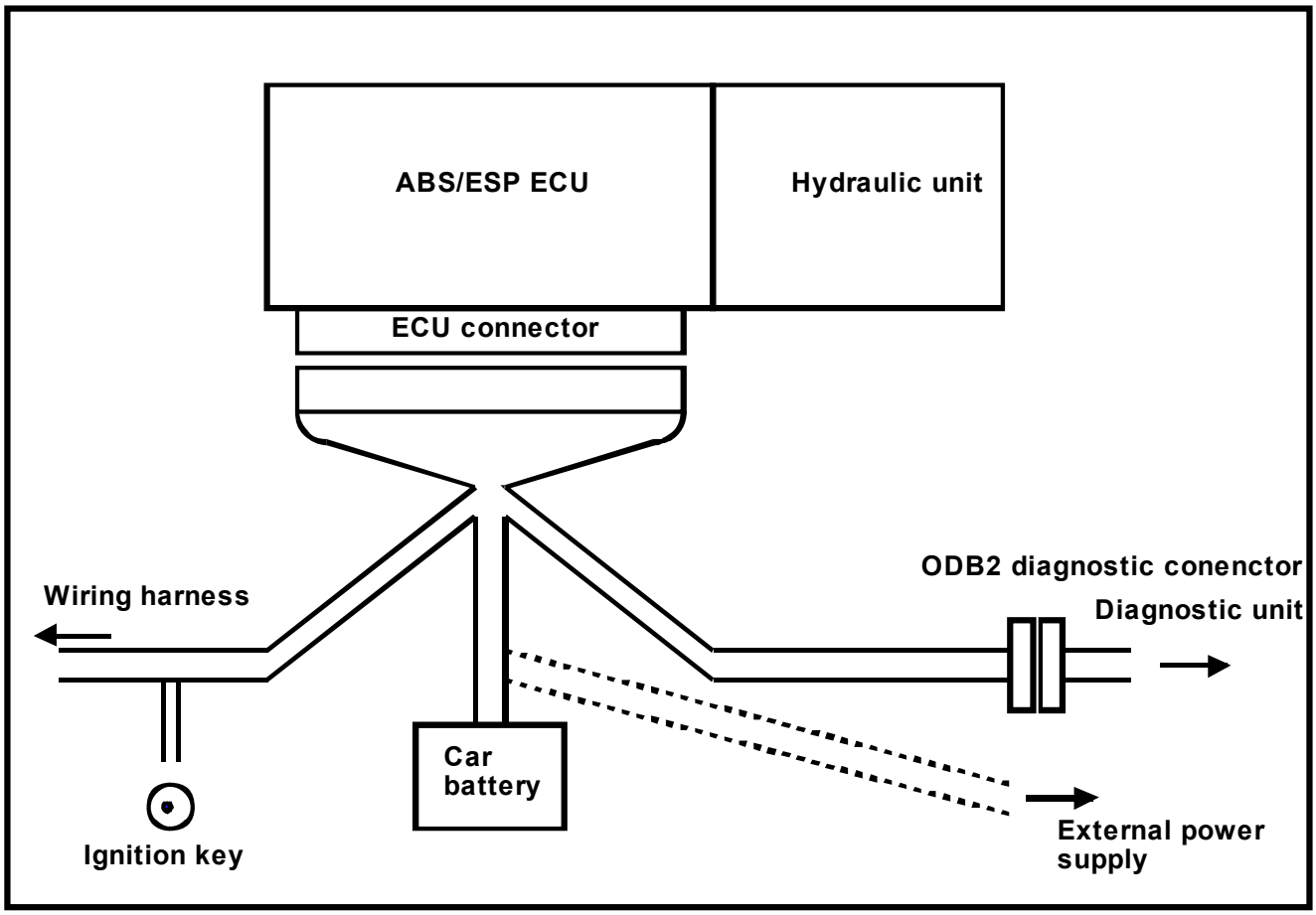
9 Electrical adaptations 电气连接

9.1 Connection at the diagnostic connector (OBD2) within the vehicle 连接车内 OBD2 接口

Handling 操作	
Operator 操作员	**Remarks 备注:**
☞ **Adapt diagnostic connector** 连接诊断接头 ☞ **Adapt power supply 连接电源** ☞ **Switch ignition key to position on** 打开点火开关	**Either in the passenger or engine compartment.** 在驾驶室或发动机舱内 **External power supply is required either to supply the car battery or the ABS/ESP ECU itself (e.g. car battery is not installed prior this process step).** 需要外接电源为汽车蓄电池或 **ABS/ESP ECU** 供应电能（例如汽车蓄电池此时尚未安装）。 **To switch on and initialize the ABS/ESP system.** 点火并初始化 **ABS/ESP** 系统

Notes 注意:
The length of the diagnostic connection wires should be covered and should not exceed 10 meters.
诊断连接线外部应当裹好，线缆长度不能超过 10 米。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

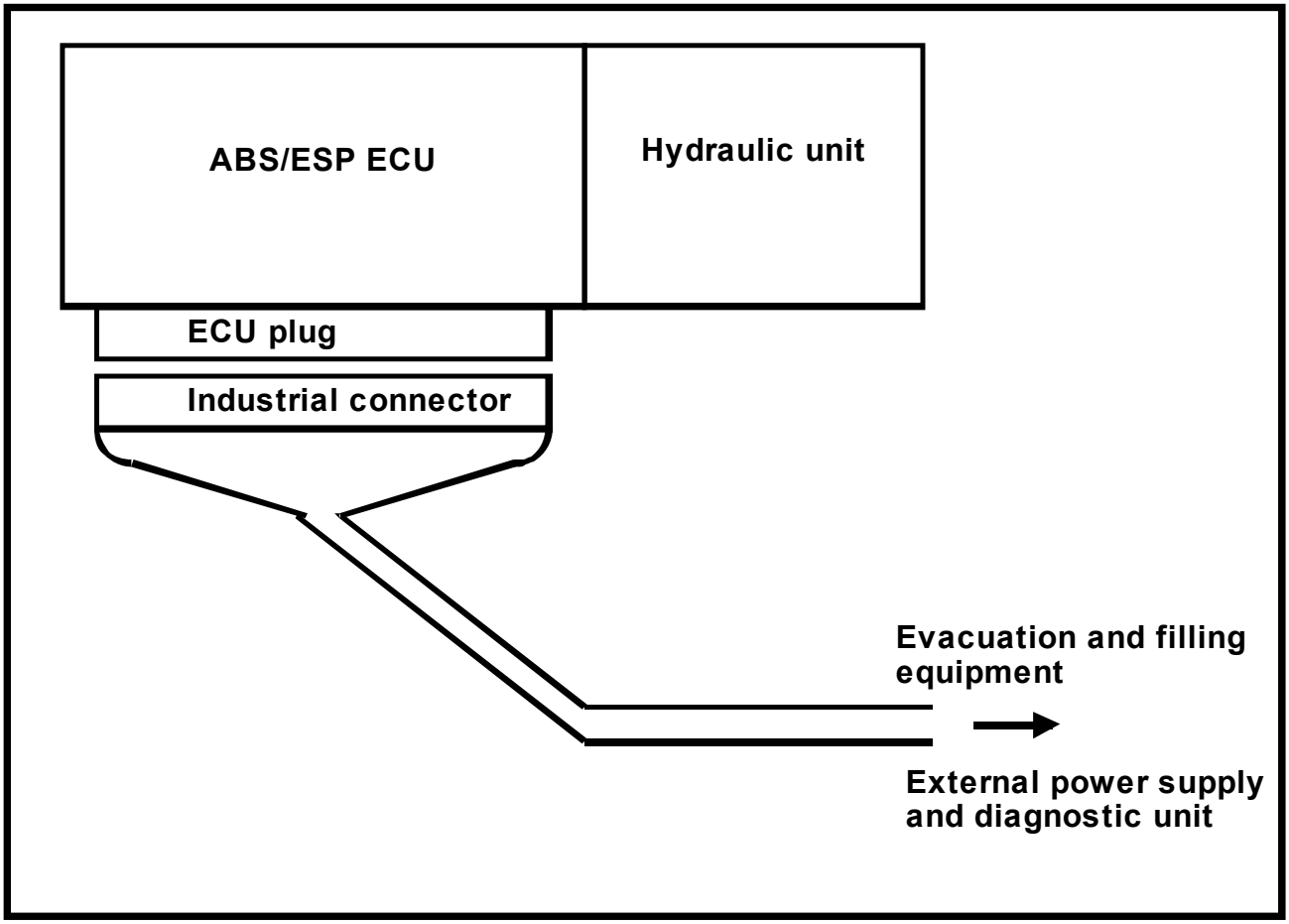
9.2 Connection at the ECU plug 连接 ECU 插头

Handling 操作	
Operator 操作员	**Remarks 备注:**
☞ **Adapt the industrial connector at the ABS/ESP ECU plug 将工业型连接器安装到 ABS/ESP ECU 插头上** **Notes:注意：** **The connector might not be connected with an electrical load respectively a switched on power supply!请不要在通电的情况下安装连接器！**	**A safe connection between the industrial connector and the ABS/ESP ECU plug is absolute required! 工业型接头和 ABS/ESP ECU 插头之间必须是安全的连接。** **The external power supply and diagnostic lines are within one connector.外接电源线和诊断线位于同一连接器中。**

Notes 注意:

The length of the diagnostic connection wires should be covered and should not exceed 10 meters.
诊断连接线外部应当裹好，线缆长度不能超过 10 米。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

10 Equipment information 设备信息

Diagnostic information 诊断信息

A diagnostic unit is required to activate the valves and the pump motor. This is required to evacuate and fill the secondary circuits of the ABS/ESP system.在对第二回路抽真空加注时，诊断仪单元应启动 ABS/ESP 的阀和泵马达。

To avoid problems with the communication (especially in case of k-line diagnosis) between the diagnostic unit and ABS/ESP ECU, the diagnostic unit should be located as close as possible at the vehicle.为了避免诊断单元和 ABS/ESP ECU 之间的通讯问题（特别是使用 k 线诊断时），诊断仪应当尽量靠近车辆。

Process bytes 过程字节

The final definition of the process byte data which should be written at the evacuation and filling process must be defined by the customer.

过程字节必须由客户定义并在抽真空加注结束后写入。

Editor menu 编辑菜单

The ECU identification data might be changed by software update during a running serial production.

To have the possibility to add these new ECU data at the evacuation and filling equipment, the editor menu must be designed in a way that a complement of these missing new ECU identification data is easily possible for the operator.

在量产过程中，ECU 识别数据可能随软件升级而发生变更。为了方便操作员将这些新数据添加进抽真空加注设备，编辑菜单应当设计得简单易用。

Test protocol 测试协议

In case of an ok test result, a print out (”test passed”) is made.如果测试合格，打印“测试通过”。.

In case of a nok test result, a failure protocol must be printed out.如果测试不合格，打印错误协议。

This protocol must include all technical information about the failed step in order to allow a better failure analysis.为了更好的分析问题，该协议必须包括错误步骤在内的所有技术信息。

An entire test protocol of all process step results must be available on request.

测试协议应当包括所有步骤的结果。

Vehicle (power supply) conditions 车辆（电源）条件

The battery must be charged properly in order to allow a sufficient power supply to the ABS/ESP system respectively the external power supply must ensure a stabile voltage level without insufficient voltage drops across the wiring.

蓄电池应妥善充电以保证 ABS/ESP 系统能得到充足的电力供应；同时外接电源必须保证电压水平稳定，沿线缆没有电压降发生。

Electrical circuit diagrams 电路图

It is the user’s responsibility to make sure they are using the latest available electrical circuit diagram!

The attached electrical wiring diagram is a generic electrical wiring diagram. These diagrams are project respectively customer specific.

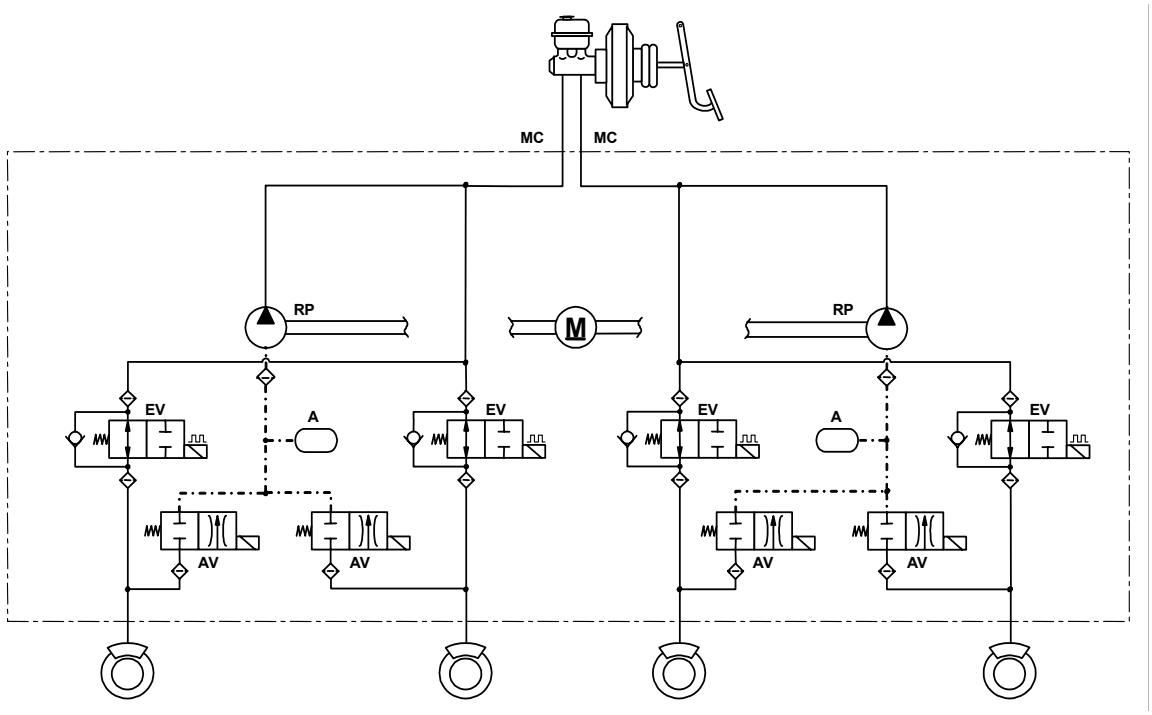
客户有责任确认自己使用的电路图为最新版本！附录中的电路图是通用版本的电路图。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

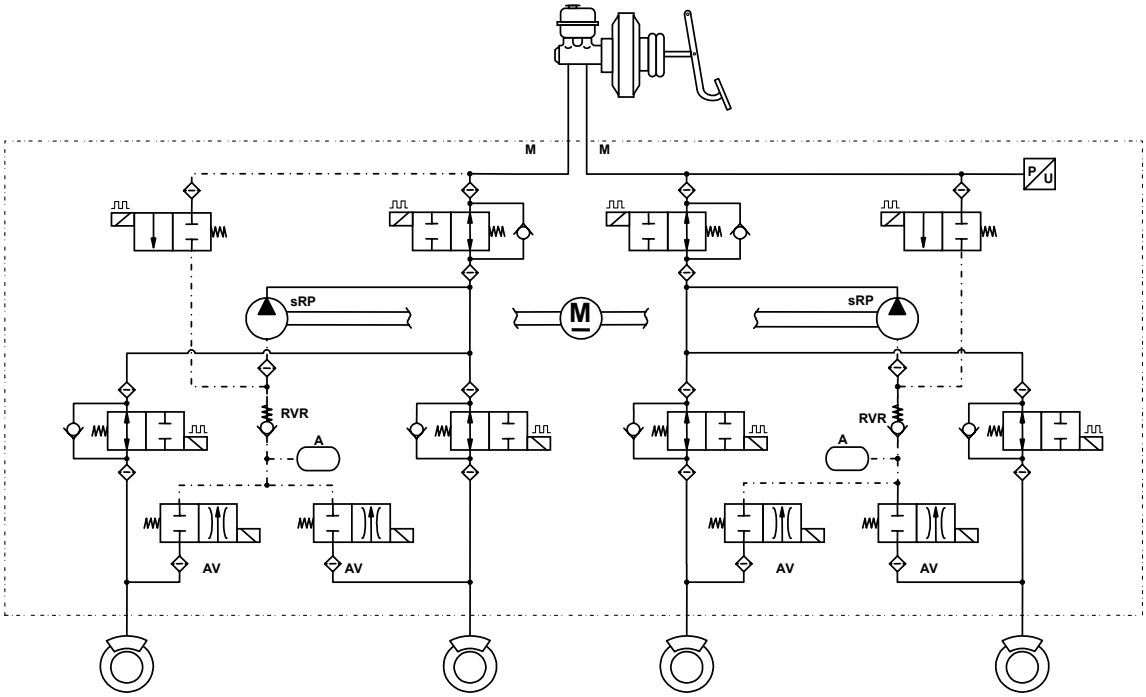
11 Attachments 附件

11.1 Hydraulic circuit diagrams 液压回路图

11.1.1 ABS 9 - independent of the brake split configuration ABS 9-适用于所有回路布置方案



11.1.2 ESP 9 - independent of the brake split configuration ESP 9-适用于所有回路布置方案

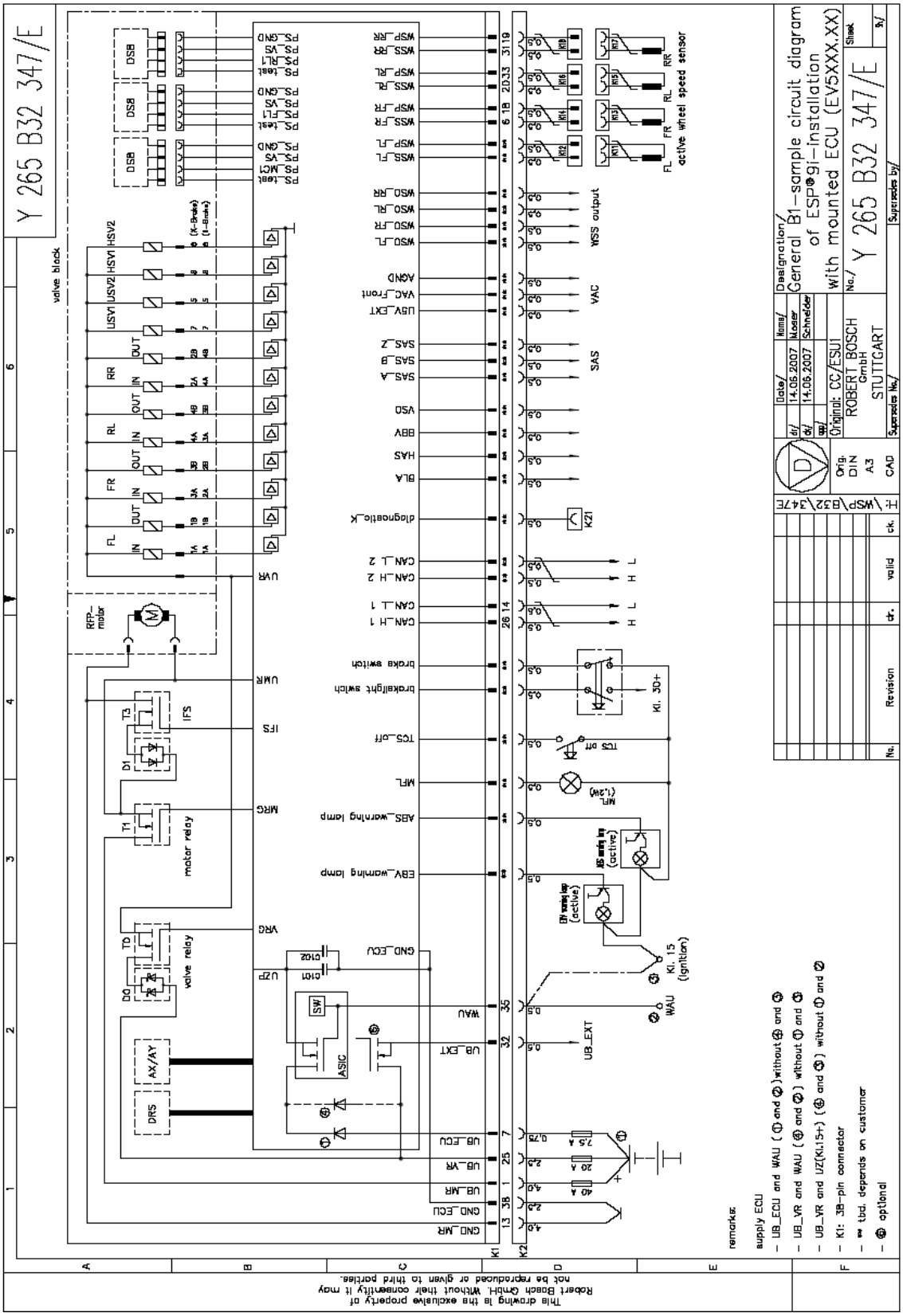


The sketched lines within the ABS respectively ESP hydraulic units show the area of the secondary and priming circuit.液压单元中虚线部分表示第二回路和吸入阀回路。

© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

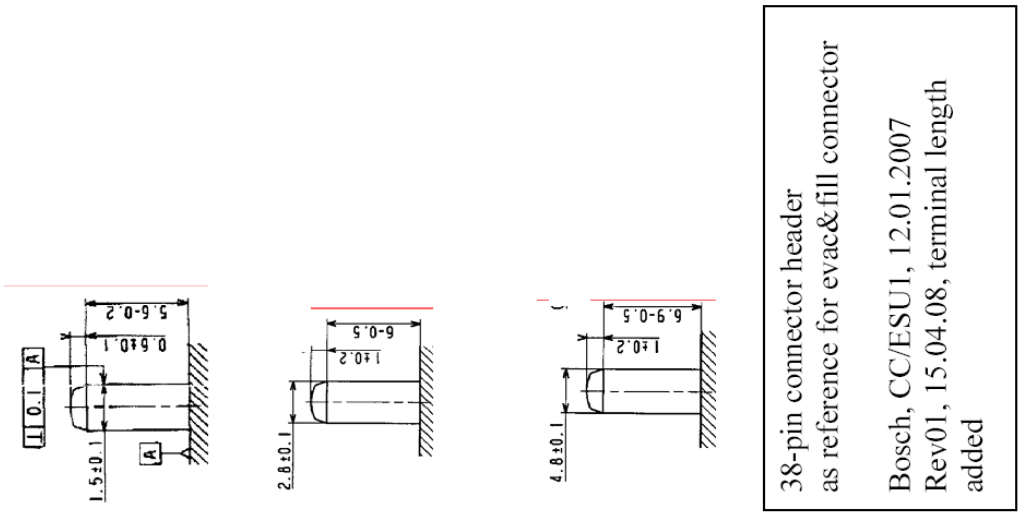
11.2 Electrical wiring diagrams 电路图

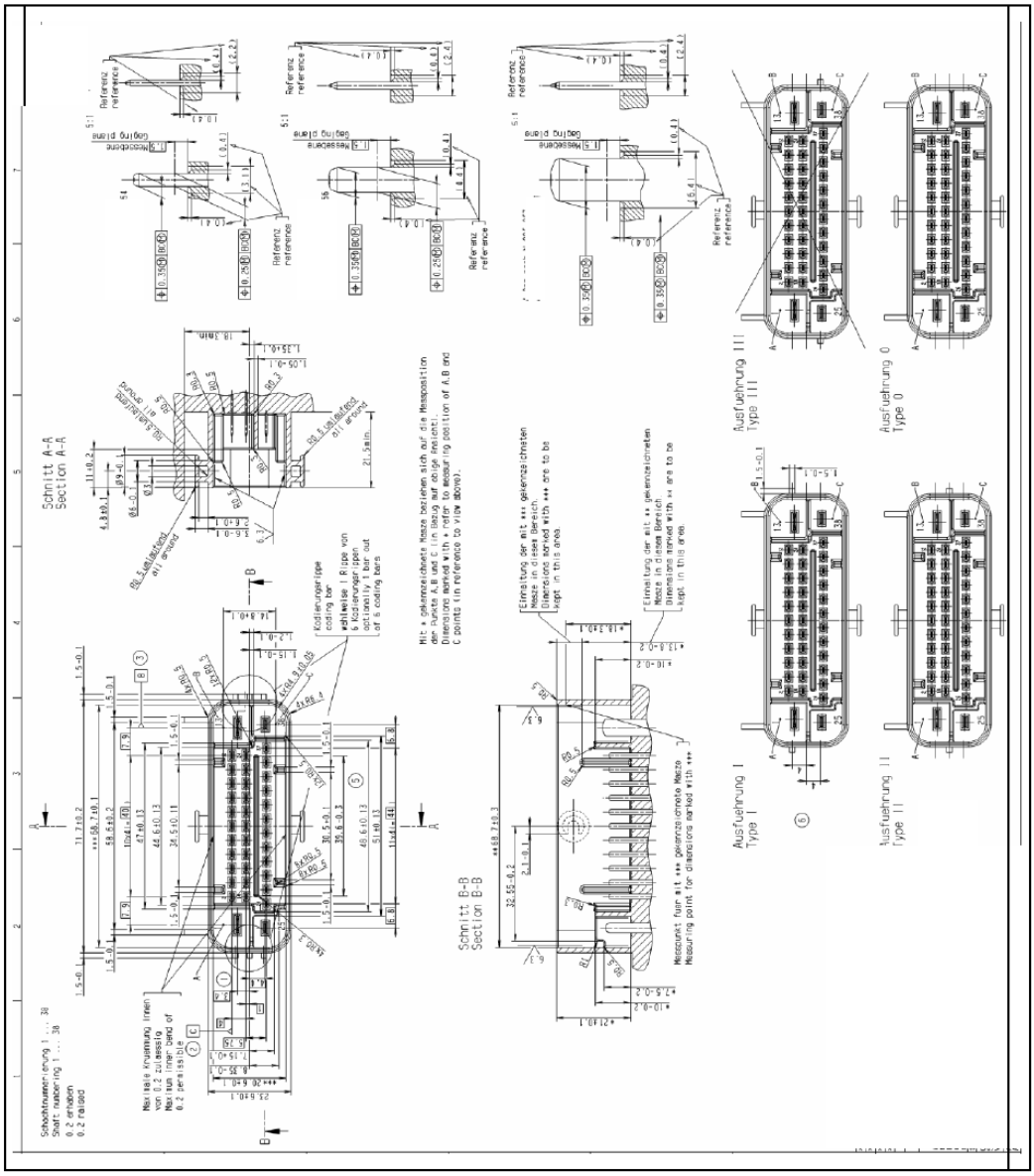
11.2.1 Electrical wiring diagram ESPi 9



© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.

11.2.2 38-pin connector system 9





© All rights with Robert Bosch GmbH, including trademark / patent applications. Any right of disposal, as copying and distribution to third parties is ours.